

Data Science and Data Analytics (WS 2025)

International Business Management (B. A.)

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This document provides the course material for Data Science and Data Analytics (B. A. – International Business Management). Upon successful completion of the course, students will be able to: recognize important technological and methodological advancements in data science and distinguish between descriptive, predictive, and prescriptive analytics; demonstrate proficiency in classifying data and variables, collecting and managing data, and conducting comprehensive data evaluations; utilize R for effective data manipulation, cleaning, visualization, outlier detection, and dimensionality reduction; conduct sophisticated data exploration and mining techniques (including PCA, Factor Analysis, and Regression Analysis) to discover underlying patterns and inform decision-making; analyze and interpret causal relationships in data using regression analysis; evaluate and organize the implementation of a data analysis project in a business environment; and communicate the results and effects of a data analysis project in a structured way.

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1 Scope and Nature of Data Science

Let's start this course with some definitions and context.

Definition of Data Science:

The field of Data Science concerns techniques for extracting knowledge from diverse data, with a particular focus on 'big' data exhibiting 'V' attributes such as volume, velocity, variety, value and veracity.

Maneth & Poulouvassilis (2016)

Definition of Data Analytics:

Data analytics is the systematic process of examining data using statistical, computational, and domain-specific methods to extract insights, identify patterns, and support decision-making. It combines competencies in data handling, analysis techniques, and domain knowledge to generate actionable outcomes in organizational contexts (Cuadrado-Gallego et al., 2023).

Definition of Business Analytics:

Business analytics is the science of posing and answering data questions related to business. Business analytics has rapidly expanded in the last few years to include tools drawn from statistics, data management, data visualization, and machine learning. There is increasing emphasis on big data handling to assimilate the advances made in data sciences. As is often the case with applied methodologies, business analytics has to be soundly grounded in applications in various disciplines and business verticals to be valuable. The bridge between the tools and the applications are the modeling methods used by managers and researchers in disciplines such as finance, marketing, and operations.

Pochiraju & Seshadri (2019)

There are many roles in the data science field, including (but not limited to):

Figure 1: **Source:** *LinkedIn*

DATA ANALYST		
Analyzing data to uncover insights and support better decision-making	SKILLS: • SQL • Excel • Power BI / Tableau • Basic Statistics	TASKS: • Creating reports & dashboards • Analyzing trends and patterns

DATA ENGINEER		
Building and maintaining data pipelines and infrastructure	SKILLS: • Python • ETL tools • Big Data • Cloud platforms	TASKS: • Cleaning and transforming raw data • Setting up data warehouses/lakes

DATA SCIENTIST		
Developing and managing data workflows and backend systems.	SKILLS: • Python / R • Machine Learning • Statistics • Data Visualization	TASKS: • Creating ML models • Predictive & prescriptive analytics

For skills and competencies required for data science activities, see [Skills Landscape](#).

1.1 Defining Data Science as an Academic Discipline

Data science emerges as an interdisciplinary field that synthesizes methodologies and insights from multiple academic domains to extract knowledge and actionable insights from data. As an academic discipline, data science represents a convergence of computational, statistical, and domain-specific expertise that addresses the growing need for data-driven decision-making in various sectors.

Data science draws from and interacts with multiple foundational disciplines:

- **Informatics / Information Systems:**

Informatics provides the foundational understanding of information processing, storage, and retrieval systems that underpin data science infrastructure. It encompasses database design, data modeling, information architecture, and system integration principles essential for managing large-scale data ecosystems. Information systems contribute knowledge about organizational data flows, enterprise architectures, and the sociotechnical aspects of data utilization in business contexts.

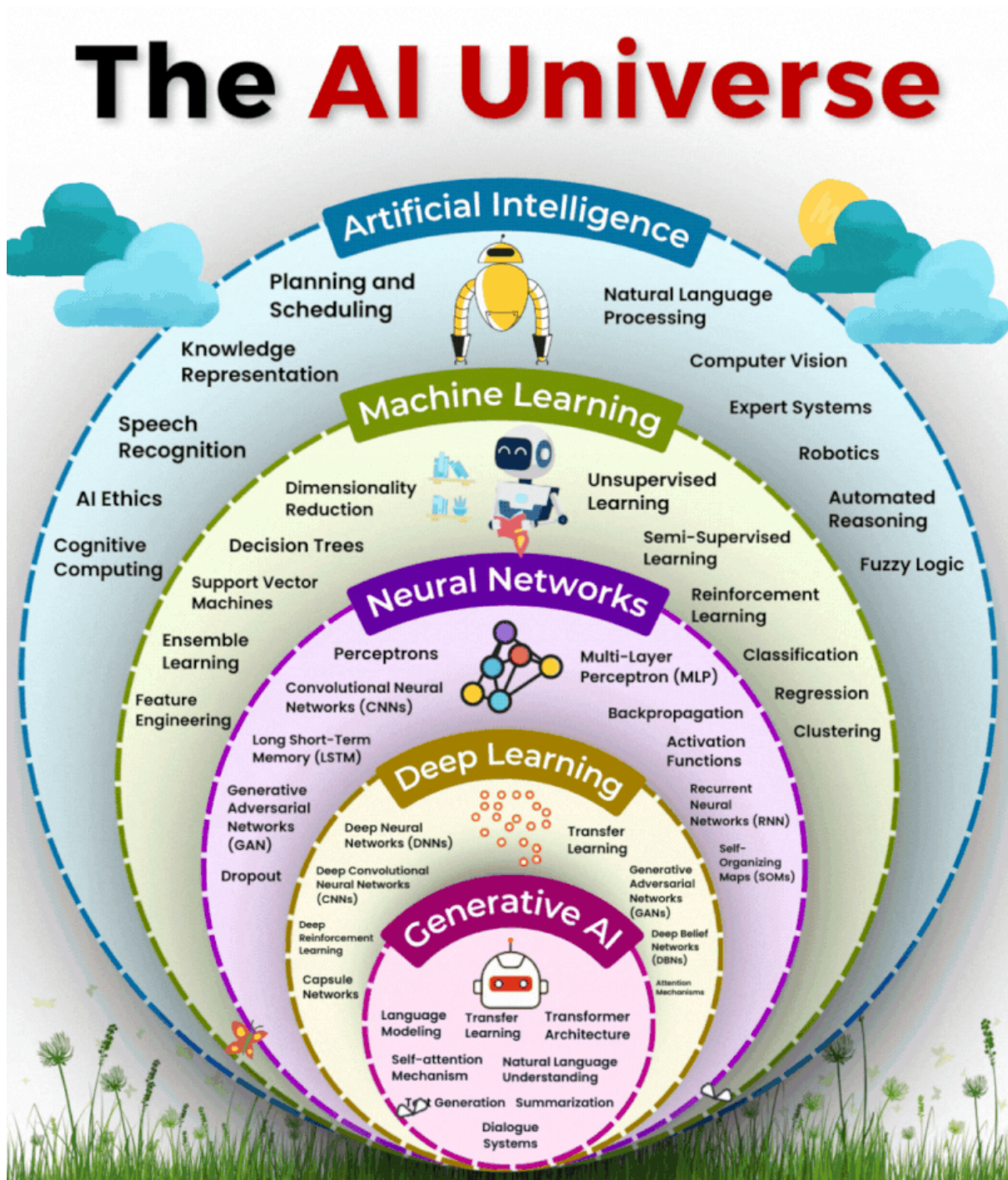
See the [Technical Applications & Data Analytics coursebook](#) by Gross (2021) for further reading on foundations in informatics.

- **Computer Science (algorithms, data structures, systems design):**

Computer science provides the computational foundation for data science through algorithm design, complexity analysis, and efficient data structures. Core contributions include machine learning algorithms, distributed computing paradigms, database systems, and software engineering practices. System design principles enable scalable data processing architectures, while computational thinking frameworks guide algorithmic problem-solving approaches essential for data-driven solutions.

See also: [Analytical Skills for Business - 1 Introduction](#) and take a look at the AI Universe overview graphic:

Figure 2: Source: *LinkedIn*



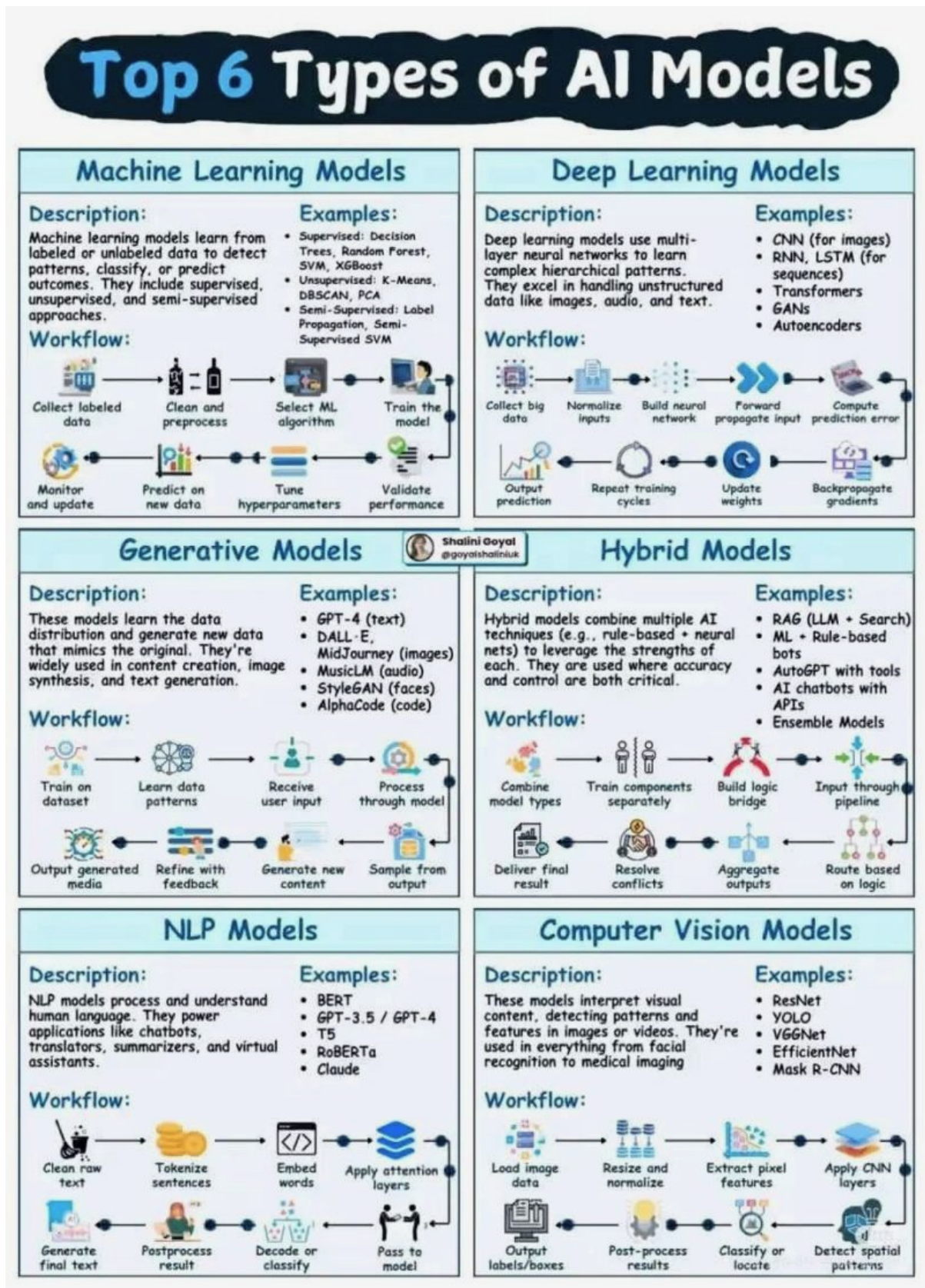
See the [Overview on no-code and low-code tools for data analytics](#) for an overview on no-code and low-code tools for data analytics and AI tooling.

- **Mathematics (linear algebra, calculus, optimization):**

Mathematics provides the theoretical backbone for data science through linear algebra (matrix operations, eigenvalues, vector spaces), calculus (derivatives, gradients, optimization), and discrete mathematics (graph theory, combinatorics). These mathematical foundations enable dimensionality reduction techniques, gradient-based optimization algorithms, statistical modeling, and the rigorous formulation of **machine learning problems** (see the

figure below). Mathematical rigor ensures the validity and interpretability of analytical results.

Figure 3: Source: *LinkedIn*



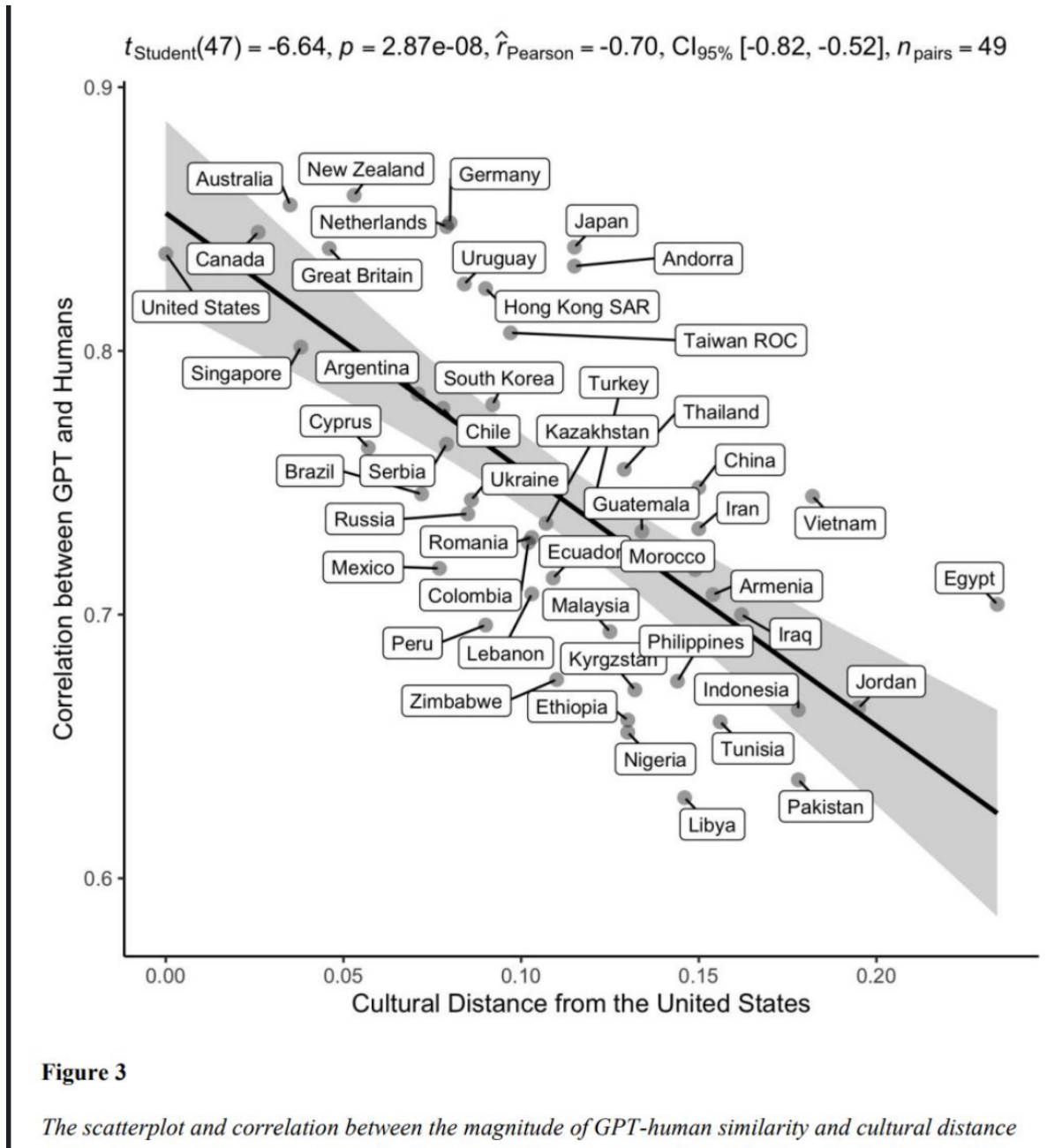
- **Statistics & Econometrics (inference, modeling, causal analysis):**

Statistics provides the methodological framework for data analysis through hypothesis testing, confidence intervals, regression analysis, and experimental design. Econometrics contributes advanced techniques for causal inference, time series analysis, and handling observational data challenges such as endogeneity and selection bias. These disciplines ensure rigorous uncertainty quantification, model validation, and the ability to draw reliable conclusions from data while understanding limitations and assumptions.

- **Social Science & Behavioral Sciences (contextual interpretation, experimental design):**

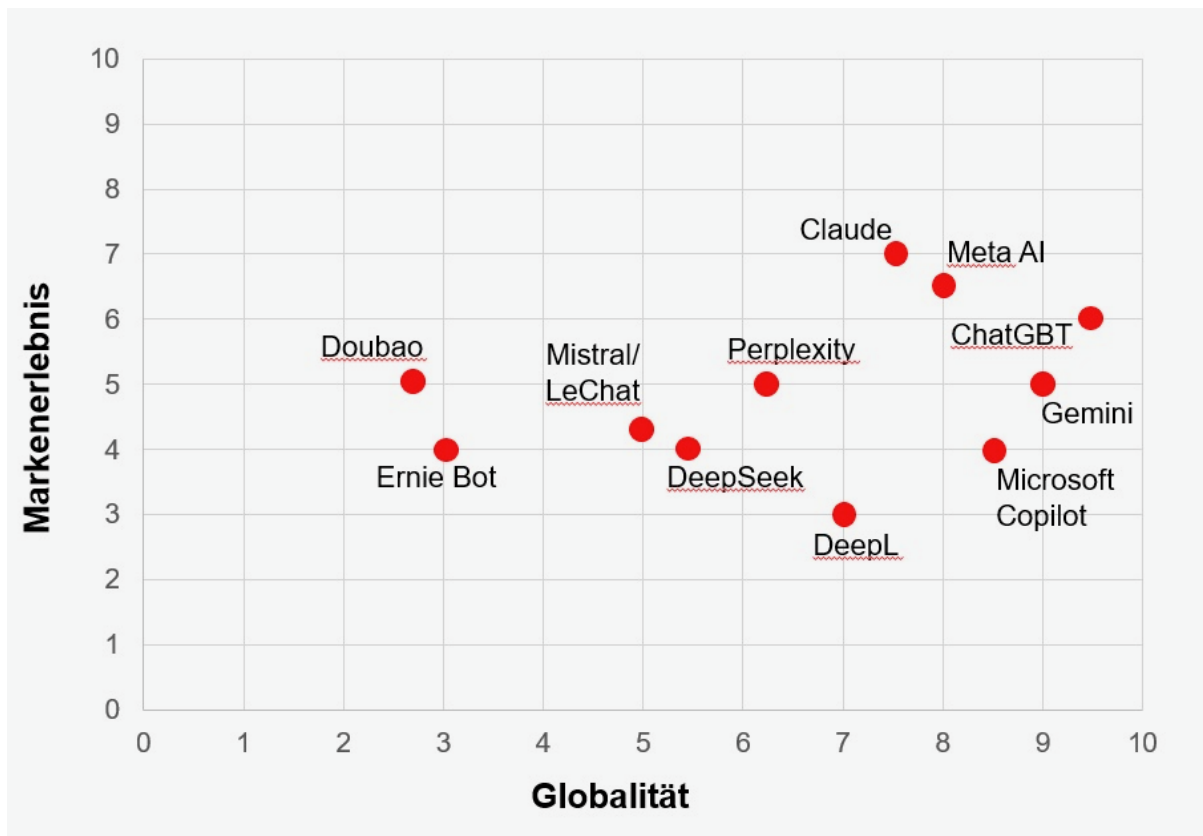
Social and behavioral sciences contribute essential understanding of human behavior, organizational dynamics, and contextual factors that influence data generation and interpretation. These disciplines provide expertise in experimental design, survey methodology, ethical considerations, and the social implications of data-driven decisions. They ensure that data science applications consider human factors, cultural context, and societal impact while maintaining ethical standards in data collection and analysis.

Figure 4: **Source:** *LinkedIn*



A recent [Business Pundit](#) study reveals that AI systems like ChatGPT, Claude, and Meta AI are developing genuine brand characteristics through their consistent tonality, personality, and emotional resonance with users. The research shows US-based AI models dominate in both global presence and emotional connection, while European systems remain functionally strong but lack distinctive brand identity. This transformation marks a shift where AI branding emerges in real-time through every interaction, making personality a core product quality and positioning systems like Claude and Meta AI as potential “superbrands” of the next decade.

Figure 5: **Source:** *Business Punk*



The interdisciplinary nature of data science requires practitioners to develop competencies across these domains while maintaining awareness of how different methodological traditions complement and inform each other. This multidisciplinary foundation enables data scientists to approach complex problems with both technical rigor and contextual understanding, ensuring that analytical solutions are both technically sound and practically relevant.

For further reading on the academic foundations of data science, see the comprehensive analysis in [Defining Data Science as an Academic Discipline](#).

1.2 Significance of Business Data Analysis for Decision-Making

Business data analysis has evolved from a supporting function to a critical strategic capability that fundamentally transforms how organizations make decisions, allocate resources, and compete in modern markets. The systematic application of analytical methods to business data enables evidence-based decision-making that reduces uncertainty, improves operational efficiency, and creates sustainable competitive advantages.

1.2.1 Strategic Decision-Making Framework

Business data analysis provides a structured approach to strategic decision-making through multiple analytical dimensions:

- **Evidence-Based Strategic Planning:** Data analysis supports long-term strategic decisions by providing empirical evidence about market trends, competitive positioning,

and organizational capabilities. Statistical analysis of historical performance data, market research, and competitive intelligence enables organizations to formulate strategies grounded in quantifiable evidence rather than intuition alone.

- **Risk Assessment and Mitigation:** Advanced analytical techniques enable comprehensive risk evaluation across operational, financial, and strategic dimensions. Monte Carlo simulations, scenario analysis, and predictive modeling help organizations quantify potential risks and develop contingency plans based on probabilistic assessments of future outcomes.
- **Resource Allocation Optimization:** Data-driven resource allocation models leverage optimization algorithms and statistical analysis to maximize return on investment across different business units, projects, and initiatives. Linear programming, integer optimization, and multi-criteria decision analysis provide frameworks for allocating limited resources to achieve optimal organizational outcomes.

1.2.2 Operational Decision Support

At the operational level, business data analysis transforms day-to-day decision-making through real-time insights and systematic performance measurement:

- **Performance Measurement and Continuous Improvement:** Key Performance Indicators (KPIs) and statistical process control methods enable organizations to monitor operational efficiency, quality metrics, and customer satisfaction in real-time. Time series analysis, control charts, and regression analysis identify trends, anomalies, and improvement opportunities that drive continuous operational enhancement.
- **Forecasting and Demand Planning:** Statistical forecasting models using techniques such as ARIMA, exponential smoothing, and machine learning algorithms enable accurate demand prediction for inventory management, capacity planning, and supply chain optimization. These analytical approaches reduce uncertainty in operational planning while minimizing costs associated with overstock or stockouts.
- **Customer Analytics and Personalization:** Advanced customer analytics leverage segmentation analysis, predictive modeling, and behavioral analytics to understand customer preferences, predict churn, and optimize retention strategies. Clustering algorithms, logistic regression, and recommendation systems enable personalized customer experiences that increase satisfaction and loyalty.

1.2.3 Tactical Decision Integration

Business data analysis bridges strategic planning and operational execution through tactical decision support:

- **Pricing Strategy Optimization:** Price elasticity analysis, competitive pricing models, and revenue optimization techniques enable dynamic pricing strategies that maximize profitability while maintaining market competitiveness. Regression analysis, A/B testing, and econometric modeling provide empirical foundations for pricing decisions.
- **Market Intelligence and Competitive Analysis:** Data analysis transforms market research and competitive intelligence into actionable insights through statistical analysis of market trends, customer behavior, and competitive positioning. Multivariate analysis,

factor analysis, and time series forecasting identify market opportunities and competitive threats.

- **Financial Performance Analysis:** Financial analytics encompassing ratio analysis, variance analysis, and predictive financial modeling enable organizations to assess financial health, identify cost reduction opportunities, and optimize capital structure decisions. Statistical analysis of financial data supports both internal performance evaluation and external stakeholder communication.

1.2.4 Contemporary Analytical Capabilities

Modern business data analysis capabilities extend traditional analytical methods through integration of advanced technologies and methodologies:

- **Real-Time Analytics and Decision Support:** Stream processing, event-driven analytics, and real-time dashboards enable immediate response to changing business conditions. Complex event processing and real-time statistical monitoring support dynamic decision-making in fast-paced business environments.
- **Predictive and Prescriptive Analytics:** Machine learning algorithms, neural networks, and optimization models enable organizations to not only predict future outcomes but also recommend optimal actions. These advanced analytical capabilities support automated decision-making and strategic scenario planning.
- **Data-Driven Innovation:** Analytics-driven innovation leverages data science techniques to identify new business opportunities, develop innovative products and services, and create novel revenue streams. Advanced analytics enable organizations to discover hidden patterns, correlations, and insights that drive innovation and competitive differentiation.

The significance of business data analysis for decision-making extends beyond technical capabilities to encompass organizational transformation, cultural change, and strategic competitive positioning. Organizations that successfully integrate analytical capabilities into their decision-making processes achieve superior performance outcomes, enhanced agility, and sustainable competitive advantages in increasingly data-driven markets.

For comprehensive coverage of business data analysis methodologies and applications, see [Advanced Business Analytics](#) and the analytical foundations outlined in Evans (2020).

For open access resources, visit [Kaggle](#), a platform for data science competitions and datasets.

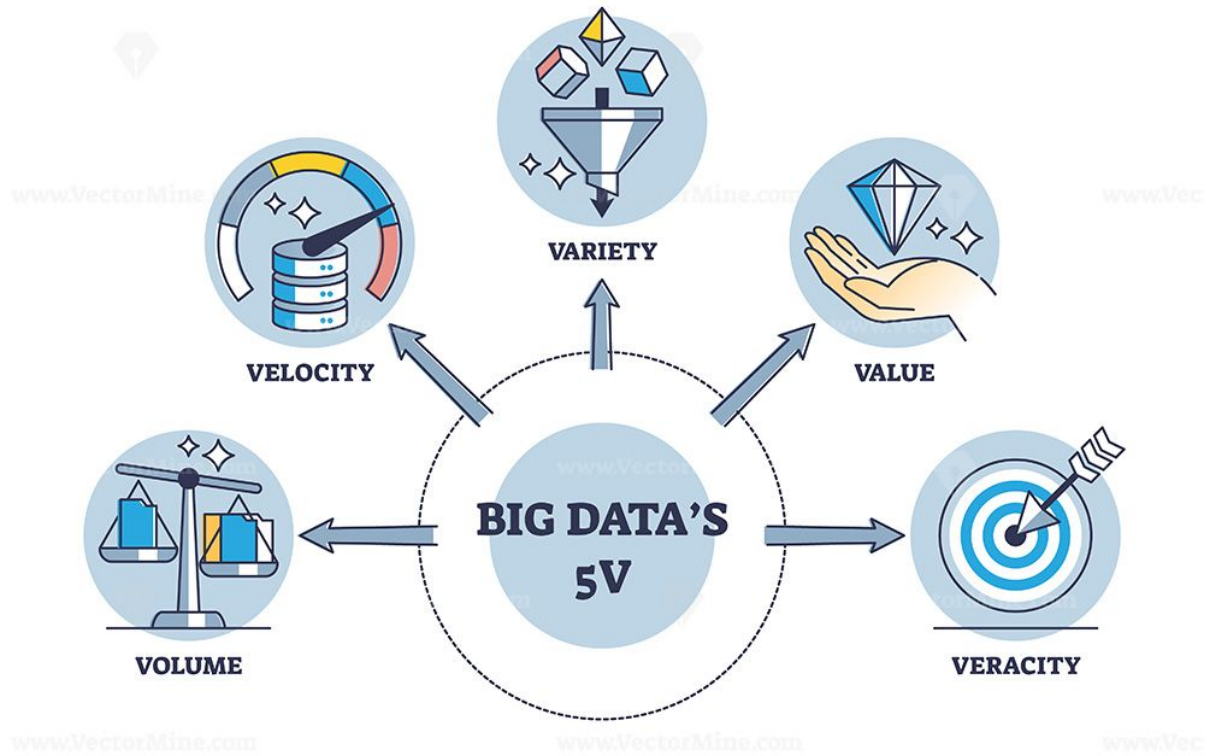
1.3 Emerging Trends

Key technological and methodological developments shaping the data landscape:

- Evolution of computing and data processing architectures.
- Digitalization of processes and platforms.
- Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL).

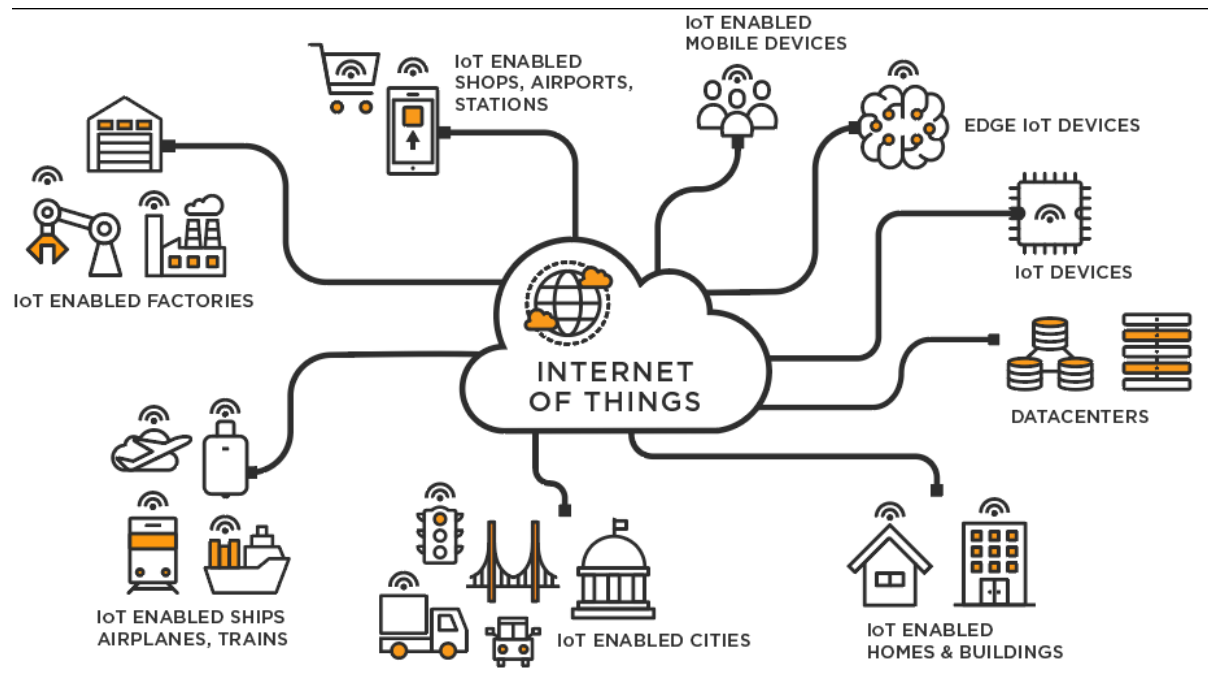
- Big Data ecosystems (volume, velocity, variety, veracity, value).

Figure 6: **Source:** *LinkedIn*



- Internet of Things (IoT) and sensor-driven data generation.

Figure 7: **Source:** <https://businesstech.bus.umich.edu/uncategorized/tech-101-internet-of-things/>

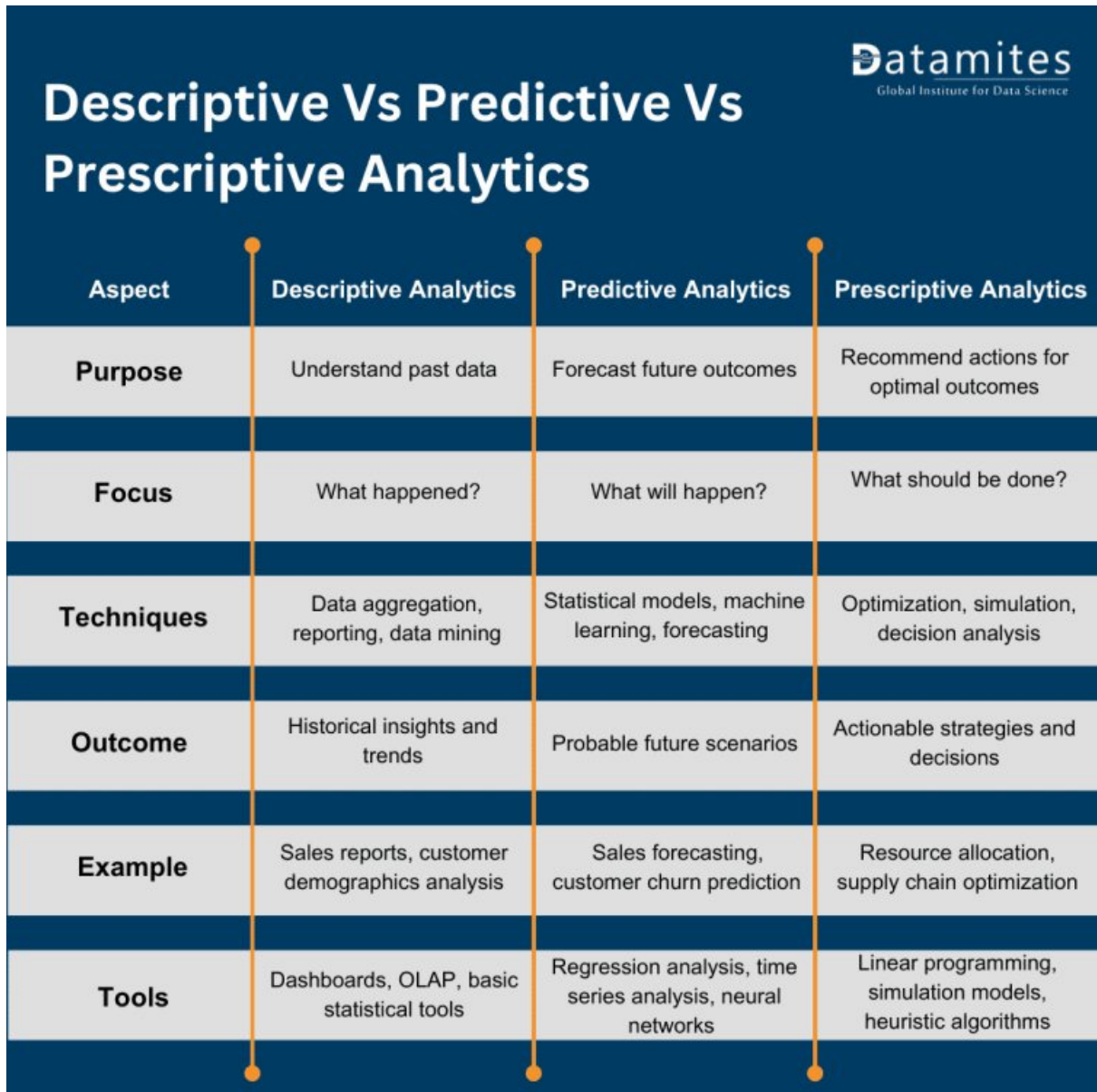


- Cloud computing and elastic infrastructure.
- Blockchain for distributed trust and data integrity.
- Industry 4.0: cyber-physical systems and automation.
- Remote and hybrid working environments: collaboration, distributed analytics, governance.

1.4 Types of Analytics

- Descriptive Analytics: What happened?
- Predictive Analytics: What is likely to happen?
- Prescriptive Analytics: What should we do?

Figure 8: **Source:** <https://datamites.com/blog/descriptive-vs-predictive-vs-prescriptive-analytics/>



Descriptive Vs Predictive Vs Prescriptive Analytics 			
Aspect	Descriptive Analytics	Predictive Analytics	Prescriptive Analytics
Purpose	Understand past data	Forecast future outcomes	Recommend actions for optimal outcomes
Focus	What happened?	What will happen?	What should be done?
Techniques	Data aggregation, reporting, data mining	Statistical models, machine learning, forecasting	Optimization, simulation, decision analysis
Outcome	Historical insights and trends	Probable future scenarios	Actionable strategies and decisions
Example	Sales reports, customer demographics analysis	Sales forecasting, customer churn prediction	Resource allocation, supply chain optimization
Tools	Dashboards, OLAP, basic statistical tools	Regression analysis, time series analysis, neural networks	Linear programming, simulation models, heuristic algorithms

2 Data Analytic Competencies

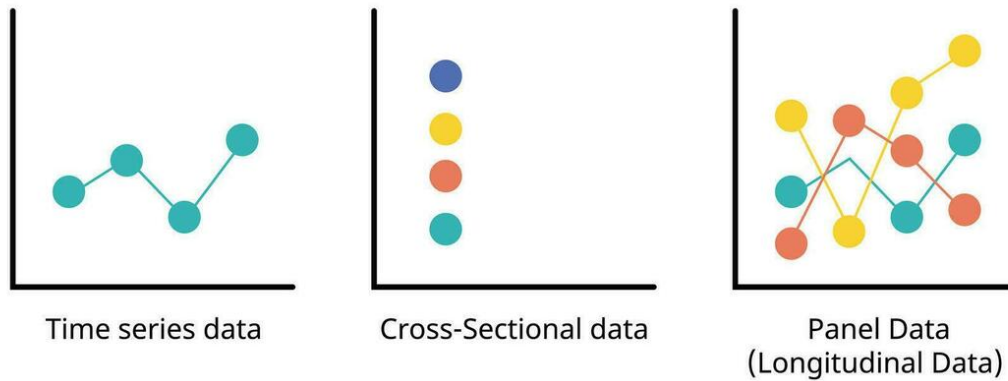
Data analytic competencies encompass the ability to apply machine learning, data mining, statistical methods, and algorithmic approaches to extract meaningful patterns, insights, and predictions from complex datasets. They include proficiency in exploratory data analysis, feature engineering, model selection, evaluation, and validation. These skills ensure rigorous interpretation of data, support evidence-based decision-making, and enable the development of robust analytical solutions adaptable to diverse health, social, and technological contexts.

2.1 Types of Data

The structure and temporal dimension of data fundamentally influence analytical approaches and statistical methods. Understanding data types enables researchers to select appropriate modeling techniques and interpret results within proper contextual boundaries.

- **Cross-sectional data** captures observations of multiple entities (individuals, firms, countries) at a single point in time. This structure facilitates comparative analysis across units but does not track changes over time. Cross-sectional studies are particularly valuable for examining relationships between variables at a specific moment and testing hypotheses about population characteristics.
- **Time-series data** records observations of a single entity across multiple time points, enabling the analysis of temporal patterns, trends, seasonality, and cyclical behaviors. Time-series methods account for autocorrelation and temporal dependencies, supporting forecasting and dynamic modeling. This data structure is essential for economic indicators, financial markets, and environmental monitoring.
- **Panel (longitudinal) data** combines both dimensions, tracking multiple entities over time. This structure offers substantial analytical advantages by controlling for unobserved heterogeneity across entities and modeling both within-entity and between-entity variation. Panel data methods support causal inference through fixed-effects and random-effects models, difference-in-differences estimation, and dynamic panel specifications.

Sequence Analytics

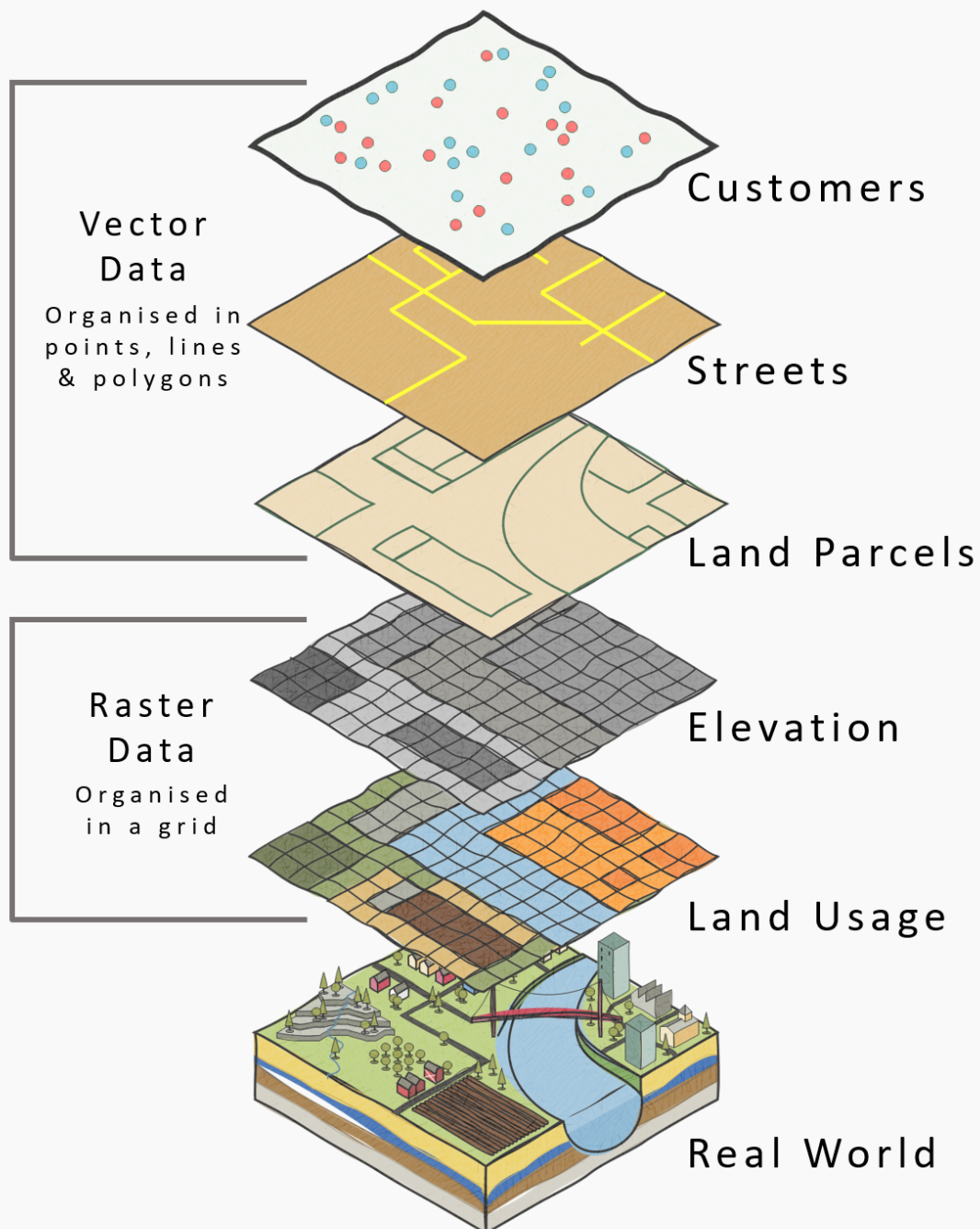


Additional data structures:

- **Geo-referenced / spatial data** is data associated with specific geographic locations, enabling spatial analysis and visualization. Techniques such as Geographic Information Systems (GIS), spatial autocorrelation, and spatial regression models are employed to analyze patterns and relationships in spatially distributed data.

What is GIS Data?

(Geographic Information Service/System)



- **Streaming / real-time data** is continuously generated data that is processed and analyzed in real-time. This data structure is crucial for applications requiring immediate insights, such as fraud detection, network monitoring, and real-time recommendation systems.

2.2 Types of Variables

- **Continuous (interval/ratio)** data is measured on a scale with meaningful intervals and a true zero point (ratio) or arbitrary zero point (interval). Examples include height, weight, temperature, and income. Continuous variables support a wide range of statistical analyses, including regression and correlation.

```
# Loading the dplyr library for data manipulation
library(dplyr)

# Assigning continuous data to a data frame and displaying it as a table
continuous_data <- data.frame(
  Height_cm = c(170, 165, 180, 175, 160, 185, 172, 168, 178, 182),
  Weight_kg = c(70, 60, 80, 75, 55, 90, 68, 62, 78, 85)
)

# Displaying the original table
print(continuous_data)
```

	Height_cm	Weight_kg
1	170	70
2	165	60
3	180	80
4	175	75
5	160	55
6	185	90
7	172	68
8	168	62
9	178	78
10	182	85

```
# Ordering the data frame by Height_cm
ordered_data <- continuous_data %>%
  arrange(Height_cm)

# Displaying the ordered data frame
print(ordered_data)
```

	Height_cm	Weight_kg
1	160	55
2	165	60
3	168	62
4	170	70
5	172	68
6	175	75
7	178	78
8	180	80
9	182	85
10	185	90


```
# Practicing:
# 1. Assign continuous data to a data frame and displaying it as a table
# 2. Order the data frame by a specific column
```

- **Count** data represents the number of occurrences of an event or the frequency of a particular characteristic. Count variables are typically non-negative integers and can be analyzed using Poisson regression or negative binomial regression.

```
# Assigning count data to a data frame and displaying it as a table
count_data <- data.frame(
  Height = c(170, 165, 182, 175, 165, 175, 175, 168, 175, 182),
  Weight = c(70, 60, 80, 75, 55, 90, 68, 62, 78, 85)
)

# Displaying the original data as a table
print(count_data)
```

	Height	Weight
1	170	70
2	165	60
3	182	80
4	175	75
5	165	55
6	175	90
7	175	68
8	168	62
9	175	78
10	182	85

```
# Ordering the data frame by Height_cm in ascending order
ordered_count_data <- count_data %>%
  arrange(desc(Height), Weight) %>%
  count(Height)

# Displaying the ordered count data
print(ordered_count_data)
```

	Height	n
1	165	2
2	168	1
3	170	1
4	175	4
5	182	2

```
# Practicing:
# 1. Assign count data to a data frame and displaying it as a table
```

- **Ordinal** data represents categories with a meaningful order or ranking but no consistent interval between categories. Examples include survey responses (e.g., Likert scales) and socioeconomic status (e.g., low, medium, high). Ordinal variables can be analyzed using non-parametric tests or ordinal regression.

```
# Loading the dplyr library for data manipulation
library(dplyr)

# Assigning ordinal data to a data frame and displaying it as a table
ordinal_data <- data.frame(
  Response = c("Strongly Disagree", "Disagree", "Neutral", "Agree", "Strongly Agree"),
  Value = c(5, 4, 3, 2, 1)
)

# Displaying the original table
print(ordinal_data)
```

	Response	Value
1	Strongly Disagree	5
2	Disagree	4
3	Neutral	3
4	Agree	2
5	Strongly Agree	1

```
# Ordering the data frame by Value
ordinal_data <- ordinal_data %>%
  arrange(Value)

# Displaying the ordered data frame
print(ordinal_data)
```

	Response	Value
1	Strongly Agree	1
2	Agree	2
3	Neutral	3
4	Disagree	4
5	Strongly Disagree	5

```
# Practicing:
# 1. Assign ordinal data to a data frame and displaying it as a table
# 2. Order the data frame by the ordinal value
```

- **Categorical (nominal / binary)** data represents distinct categories without any inherent order. Nominal variables have two or more categories (e.g., gender, race, or marital status), while binary variables have only two categories (e.g., yes/no, success/failure). Categorical variables can be analyzed using chi-square tests or logistic regression.

```
# Loading the dplyr library for data manipulation
library(dplyr)

# Assigning categorical data to a data frame and displaying it as a table
categorical_data <- data.frame(
  Event = c("A", "C", "B", "D", "E"),
  Category = c("Type1", "Type2", "Type1", "Type3", "Type2")
)
```

```
)

# Displaying the original table
print(categorical_data)
```

	Event	Category
1	A	Type1
2	C	Type2
3	B	Type1
4	D	Type3
5	E	Type2

```
# Ordering the data frame by Event
categorical_data <- categorical_data %>%
  arrange(Event)

# Displaying the table
print(categorical_data)
```

	Event	Category
1	A	Type1
2	B	Type1
3	C	Type2
4	D	Type3
5	E	Type2

```
# Practicing:
# 1. Assign categorical data to a data frame and displaying it as a table
# 2. Order the data frame by a specific column
```

- **Compositional or hierarchical structures** represent data with a part-to-whole relationship or nested categories. Examples include demographic data (e.g., age groups within gender) and geographical data (e.g., countries within continents). Compositional data can be analyzed using techniques such as hierarchical clustering or multilevel modeling.

```
# Loading the dplyr library for data manipulation
library(dplyr)

# Assigning hierarchical data to a data frame and displaying it as a table
hierarchical_data <- data.frame(
  Country = c("USA", "Canada", "USA", "Canada", "Mexico"),
  State_Province = c("California", "Ontario", "Texas", "Quebec", "Jalisco"),
  Population_Millions = c(39.5, 14.5, 29.0, 8.5, 8.3)
)

# Displaying the table
print(hierarchical_data)
```

	Country	State_Province	Population_Millions
1	USA	California	39.5
2	Canada	Ontario	14.5
3	USA	Texas	29.0
4	Canada	Quebec	8.5
5	Mexico	Jalisco	8.3

```
# Ordering the data frame by Country and then State_Province
hierarchical_data <- hierarchical_data %>%
  arrange(Country, State_Province)

# Displaying the ordered data frame
print(hierarchical_data)
```

	Country	State_Province	Population_Millions
1	Canada	Ontario	14.5
2	Canada	Quebec	8.5
3	Mexico	Jalisco	8.3
4	USA	California	39.5
5	USA	Texas	29.0

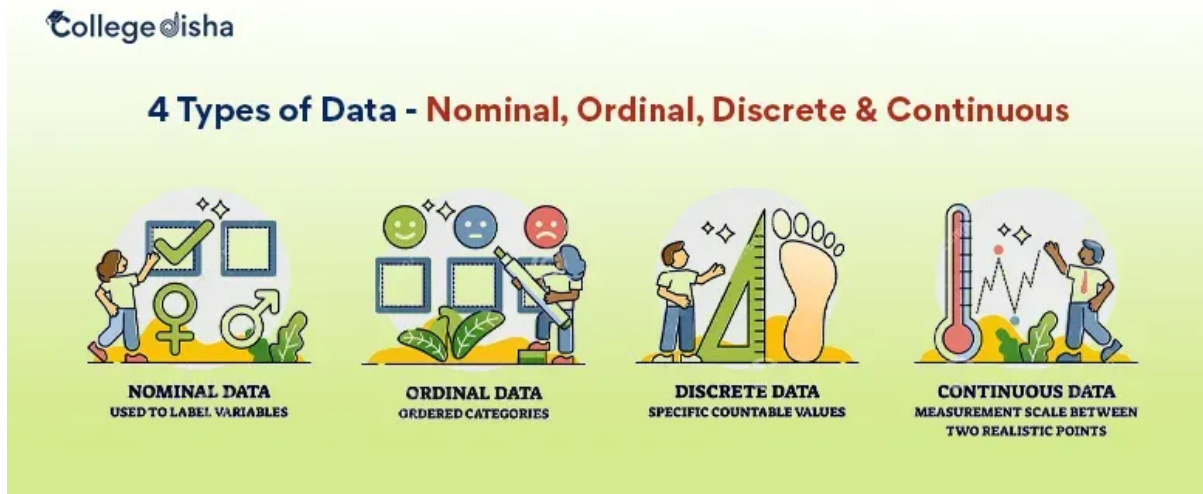
```
# Grouping the data by Country and summarizing total population
hierarchical_data_grouped <- hierarchical_data %>%
  group_by(Country) %>%
  summarise(Total_Population = sum(Population_Millions))

# Displaying the grouped data frame
print(hierarchical_data_grouped)
```

```
# A tibble: 3 x 2
  Country Total_Population
  <chr>      <dbl>
1 Canada      23
2 Mexico      8.3
3 USA       68.5
```

```
# Practicing:
# 1. Assign hierarchical data to a data frame and displaying it as a table
# 2. Order the data frame by multiple columns
```

Figure 11: Source: <https://www.collegedisha.com/>



Some small datasets to start with

A custom R package `ourdata` has been created to provide some small datasets (and also some helper R functions) for practice. You can install it from GitHub using the following commands:

```
# Installing Github R packages
devtools::install_github("DrBenjamin/ourdata")
```

Using the package and exploring its documentation:

```
# Loading package
library(ourdata)

# Opening help of package
??ourdata

# Showing welcome message
ourdata()
```

This is the R package ``ourdata`` used for Data Science courses at the Fresenius University of Applied Sciences. Type ``help(ourdata)`` to display the content. Type ``ourdata_website()`` to open the package website. Have fun in the course!
Benjamin Gross

```
# Printing some datasets
print(koelsch)
```

```
  Jahr   Koelsch
1 2017 186784000
2 2018 191308000
3 2019 179322000
4 2020 169182000
```

```
print(kirche)
```

```
  Jahr Austritte
1 2017    364711
2 2018    437416
3 2019    539509
4 2020    441390
```

```
# Using help function from the package for a specific function from the `ourdata` R package
help(combine)
```

Help on topic 'combine' was found in the following packages:

Package	Library
ourdata	/home/runner/R-library
dplyr	/home/runner/R-library

Using the first match ...

```
# Using the `combine` function from the `ourdata` R package to combine two vectors into a data frame
ourdata::combine(kirche$Jahr, koelsch$Jahr, kirche$Austritte, koelsch$Koelsch)
```

```
'data.frame':  4 obs. of  3 variables:
 $ C1: chr  "2017" "2018" "2019" "2020"
 $ C2: num  364711 437416 539509 441390
 $ C3: num  1.87e+08 1.91e+08 1.79e+08 1.69e+08
```

```
   C1    C2    C3
1 2017 364711 186784000
2 2018 437416 191308000
3 2019 539509 179322000
4 2020 441390 169182000
```

**** A messy dataset example****

We collect [data](#) from OECD (Organization for Economic Co-operation and Development), an international organization that works to build better policies for better lives. The dataset shows many columns (variables) with non-informative data and needs to be cleaned (wrangled) before analysis. First load the data into R from CSV file:

```
# Reading the dataset from a CSV file
preventable_deaths <- read.csv(
  "./topics/data/OECD_Preventable_Deaths.csv",
  stringsAsFactors = FALSE
)
```

or use the dataset directly from the `ourdata` R package:

```
# Reading the dataset from `ourdata` R package
library(ourdata)
preventable_deaths <- oecd_preventable
```

First we explore the data:

```
# Viewing structure of the dataset
str(preventable_deaths)

# Viewing first few rows
head(preventable_deaths)

# Checking dimensions
dim(preventable_deaths)

# Viewing column names
colnames(preventable_deaths)

# Summary statistics
summary(preventable_deaths)

# Checking for missing values
colSums(is.na(preventable_deaths))
```

Now we can start cleaning the data by removing non-informative columns and rows with missing values:

```
# Loading necessary library
library(dplyr)

# Selecting relevant columns for analysis
df_clean <- preventable_deaths %>%
  select(
    REF_AREA,
    Reference.area,
    TIME_PERIOD,
    OBS_VALUE
  ) %>%
  rename(
    country_code = REF_AREA,
    country = Reference.area,
    year = TIME_PERIOD,
    death_rate = OBS_VALUE
  )

# Converting year to numeric
df_clean$year <- as.numeric(df_clean$year)

# Converting death_rate to numeric (handling empty strings)
df_clean$death_rate <- as.numeric(df_clean$death_rate)
```



```
# Removing rows with missing death rates
df_clean <- df_clean %>%
  filter(!is.na(death_rate))
```

```
# Viewing cleaned data structure
str(df_clean)
```

```
'data.frame': 1162 obs. of 4 variables:
 $ country_code: chr "AUS" "AUS" "AUS" "AUS" ...
 $ country : chr "Australia" "Australia" "Australia" "Australia" ...
 $ year : num 2010 2011 2012 2013 2014 ...
 $ death_rate : num 110 109 105 105 107 108 103 103 101 104 ...
```

```
# Summary of cleaned data
summary(df_clean)
```

country_code	country	year	death_rate
Length:1162	Length:1162	Min. :2010	Min. : 34.0
Class :character	Class :character	1st Qu.:2013	1st Qu.: 75.0
Mode :character	Mode :character	Median :2016	Median :116.0
		Mean :2016	Mean :128.7
		3rd Qu.:2019	3rd Qu.:156.0
		Max. :2023	Max. :453.0

Showing some basic statistics of the cleaned data:

```
# Loading necessary library
library(dplyr)

# Printing overall statistics
cat("Mean death rate:", mean(df_clean$death_rate, na.rm = TRUE), "\n")
```

Mean death rate: 128.7151

```
cat("Median death rate:", median(df_clean$death_rate, na.rm = TRUE), "\n")
```

Median death rate: 116

```
cat("Standard deviation:", sd(df_clean$death_rate, na.rm = TRUE), "\n")
```

Standard deviation: 69.58822

```
cat("Min death rate:", min(df_clean$death_rate, na.rm = TRUE), "\n")
```

Min death rate: 34

```
cat("Max death rate:", max(df_clean$death_rate, na.rm = TRUE), "\n")
```

Max death rate: 453

```
# Printing statistics by country
country_stats <- df_clean %>%
  group_by(country) %>%
  summarise(
    mean_rate = mean(death_rate, na.rm = TRUE),
    median_rate = median(death_rate, na.rm = TRUE),
    sd_rate = sd(death_rate, na.rm = TRUE),
    min_rate = min(death_rate, na.rm = TRUE),
    max_rate = max(death_rate, na.rm = TRUE),
    n_observations = n()
  ) %>%
  arrange(desc(mean_rate))
print(country_stats)
```

A tibble: 46 x 7

	country	mean_rate	median_rate	sd_rate	min_rate	max_rate	n_observations
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
1	South Africa	330.	338	53.1	241	438	22
2	Latvia	230.	226.	65.8	151	364	28
3	Lithuania	225.	204.	67.9	134	340	28
4	Romania	223.	220.	46.5	172	303	24
5	Hungary	218.	208	72.4	141	375	28
6	Mexico	216.	217	75.5	155	453	26
7	Bulgaria	193.	186	45.5	156	378	26
8	Brazil	185.	172.	51.6	133	356	24
9	Slovak Republ~	178.	167	43.4	130	308	24
10	Estonia	172.	167	59.7	100	265	26

i 36 more rows

```
# Printing statistics by year
year_stats <- df_clean %>%
  group_by(year) %>%
  summarise(
    mean_rate = mean(death_rate, na.rm = TRUE),
    median_rate = median(death_rate, na.rm = TRUE),
    n_countries = n_distinct(country)
  ) %>%
  arrange(year)
print(year_stats)
```

A tibble: 14 x 4

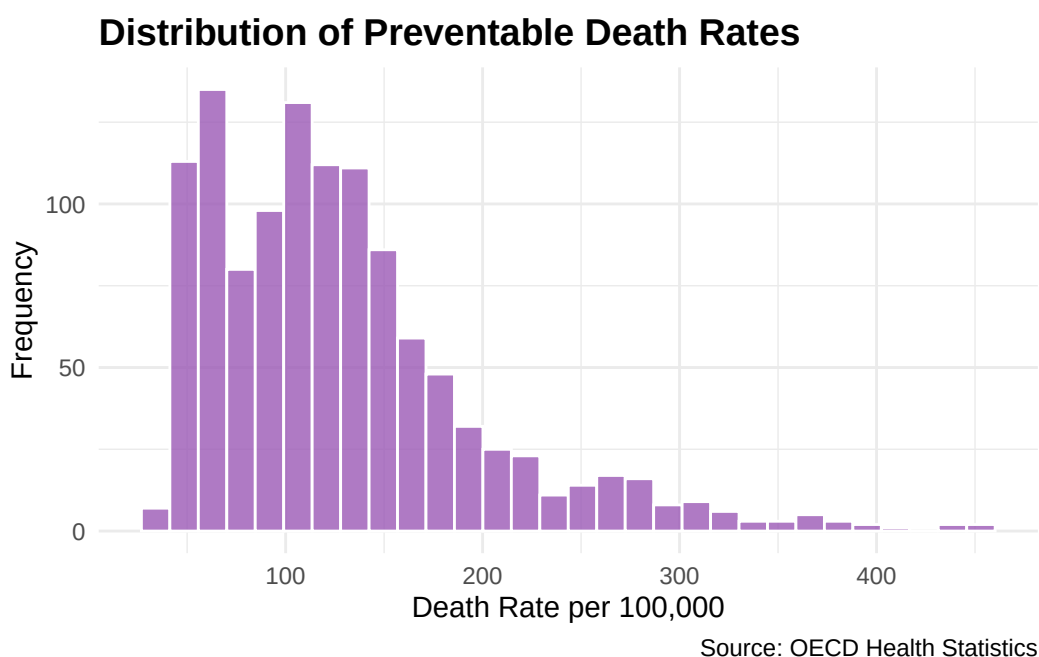
	year	mean_rate	median_rate	n_countries
	<dbl>	<dbl>	<dbl>	<int>
1	2010	141.	128	45
2	2011	135.	124.	44

3	2012	133.	120	45
4	2013	130.	116	45
5	2014	126.	117	46
6	2015	125.	114.	45
7	2016	124.	112.	46
8	2017	122.	111	45
9	2018	121.	108.	45
10	2019	118.	106.	44
11	2020	137.	118.	42
12	2021	148.	118.	40
13	2022	120.	108.	35
14	2023	112.	99.5	14

To visualize the cleaned data, we can create a distribution plot of preventable death rates:

```
# Loading necessary library
library(ggplot2)

# Plotting distribution of death rates
ggplot(df_clean, aes(x = death_rate)) +
  geom_histogram(bins = 30, fill = "#9B59B6", color = "white", alpha = 0.8) +
  labs(
    title = "Distribution of Preventable Death Rates",
    x = "Death Rate per 100,000",
    y = "Frequency",
    caption = "Source: OECD Health Statistics"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 14)
  )
```



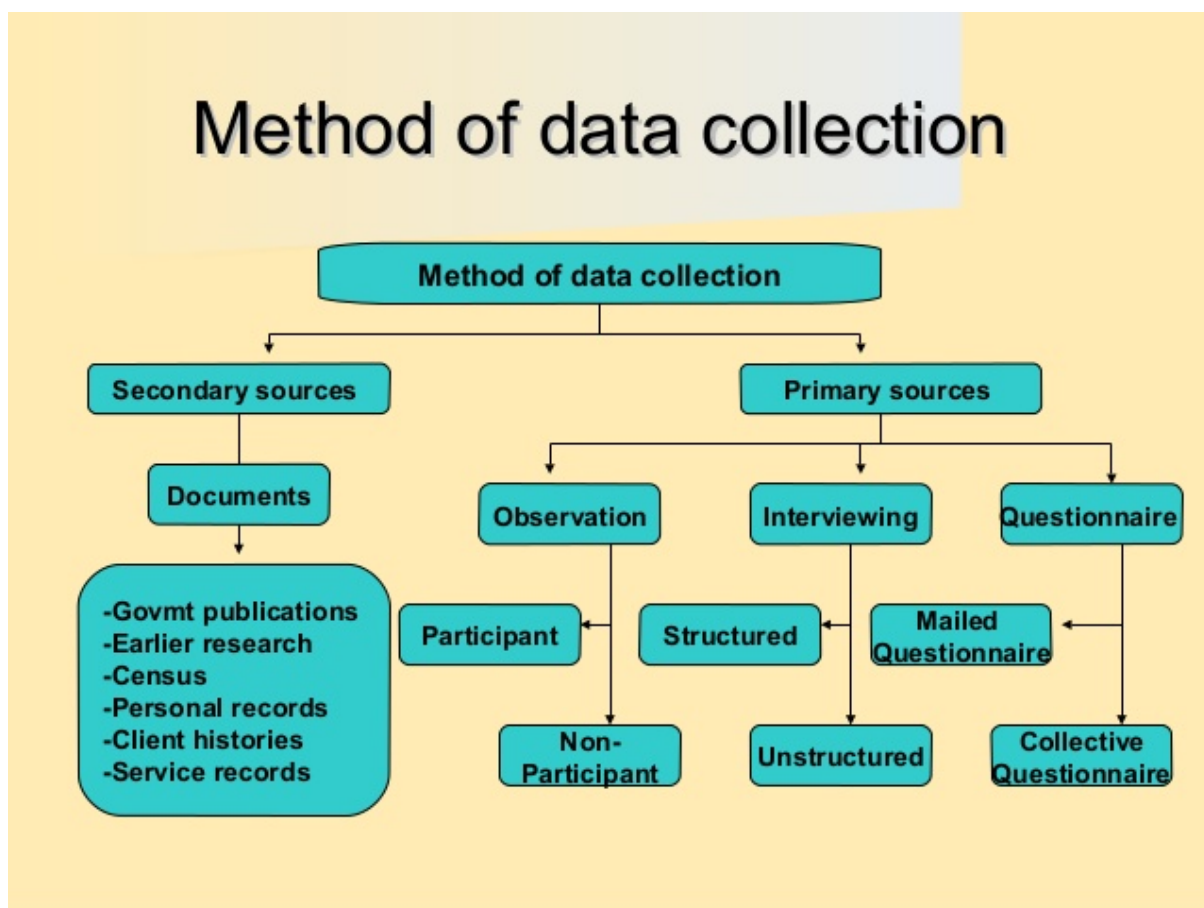
2.3 Conceptual Framework: Knowledge & Understanding of Data

- **Clarify analytical purpose and domain context** to guide data selection and interpretation.
- **Define entities, observational units, and identifiers** to ensure accurate data representation.
- **Align business concepts with data structures** for meaningful analysis.

2.4 Data Collection

Data collection forms the foundational stage of any data science project, requiring systematic approaches to gather information that aligns with research objectives and analytical requirements. As outlined in modern statistical frameworks, effective data collection strategies must balance methodological rigor with practical constraints (M. & Hardin, 2021).

Figure 12: Methods of Data Collection



2.4.1 Core Data Collection Competencies

The competencies required for effective data collection encompass both technical proficiency and methodological understanding (see [Data Collection Competencies.pdf](#)):

- **Source Identification and Assessment:** Systematically identify internal and external data sources, evaluating their relevance, quality, and accessibility for the analytical objectives.

- **Data Acquisition Methods:** Implement appropriate collection techniques including APIs (for instance see [Spotify API tutorial](#) and [Postman Spotify tutorial](#)), database queries, survey instruments, sensor networks, web scraping, and third-party vendor partnerships, ensuring methodological alignment with research design.

For the two projects below, here are some data sources recommendations (automatically created by Perplexity AI Deep Research):

- [Project car emission/sales](#)
- [Project Spotify](#)
- **Quality and Governance Framework:** Establish protocols for assessing data provenance, licensing agreements, ethical compliance, and regulatory requirements (GDPR, industry-specific standards).
- **Methodological Considerations:** Apply principles from research methodology to ensure data collection approaches support valid statistical inference and minimize bias introduction during the acquisition process.

2.4.2 Contemporary Data Collection Landscape

Modern data collection operates within an increasingly complex ecosystem characterized by diverse data types, real-time requirements, and distributed sources. The integration of traditional survey methods with emerging IoT sensors, social media APIs, and automated data pipelines requires comprehensive competency frameworks that address both technical implementation and methodological validity.

For comprehensive coverage of data collection methodologies and best practices, refer to: [Research Methodology - Data Collection](#)

2.5 Data Management

- Organize: schema design, naming conventions.
- Clean: resolve duplicates, inconsistencies, missingness.
- Convert: type casting, normalization, encoding.
- Curate: maintain lineage, documentation, metadata.
- Preserve: backups, versioning (see also [Analytical Skills for Business - Implementing version control systems](#)), retention policies.

Data Management in data science curricula requires a coherent, multi-facet framework that spans data quality, FAIR stewardship, master data governance, privacy/compliance, and modern architectures. Data quality assessment and governance define objective metrics (completeness, accuracy, consistency, plausibility, conformance) and governance processes that balance automated checks with human oversight. FAIR data principles provide a practical blueprint for metadata-rich stewardship to support findability, accessibility, interoperability, and reuse through machine-actionable metadata and persistent identifiers.

Master Data Management ensures clean, trusted core entities across systems via governance and harmonization. Data privacy, security, and regulatory compliance embed responsible data handling and risk management, guided by purpose limitation, data minimization, accuracy, storage limitations, integrity/confidentiality, and accountability. Emerging trends in cloud-native data platforms, ETL/ELT (Extract, Transform, Load or Extract, Load, Transform), data lakes/lakehouses, and broader metadata automation shape scalable storage/compute and

governance, enabling reproducible analytics and ML workflows. Together, these strands underpin trustworthy, discoverable, and compliant data inputs for research and coursework (Weiskopf & Weng, 2016; Wilkinson et al., 2016; GO FAIR Foundation, n.d.; Semarchy, n.d.; IBM, n.d.).

Tools like [dbt](#), [Apache Airflow](#), and data catalog platforms (e.g., [Alation](#), [Collibra](#)) operationalize data management practices. Also the [n8n](#) automation tool can be used for automating data workflows. See the [Analytical Skills for Business course handbook](#) on this topic. In late 2025 there was a free [n8n certification course](#) added to the n8n documentation.

2.6 Data Evaluation

- Define data quality dimensions and assessment frameworks.
- Distinguish validation from verification in data pipelines.
- Apply statistical methods and data profiling for evaluation.
- Balance automated and manual (human-in-the-loop) evaluation approaches.
- Implement tools, workflows, and governance for data evaluation.

Data evaluation ensures datasets are fit-for-purpose, reliable, and trustworthy throughout the analytical lifecycle. It encompasses systematic assessment of data quality dimensions (completeness, accuracy, consistency, validity, timeliness, uniqueness), validation and verification processes, and strategic application of statistical methods and data profiling. Quality dimensions provide measurable criteria for determining whether data meets analytical requirements, while assessment frameworks translate these into actionable metrics enabling objective measurement and contextual interpretation.

Validation and verification are complementary processes essential for data integrity. Validation checks occur at entry points, preventing bad data through constraint checks, format validation, and business rule enforcement. Verification involves post-collection checks ensuring data remains accurate and consistent over time and across system boundaries, supporting reproducibility and traceability. Together, validation acts as a gatekeeper while verification provides ongoing quality assurance.

Statistical methods form the technical foundation for evaluation. Outlier detection techniques (z-score, IQR, DBSCAN) identify anomalous observations requiring investigation. Distribution checks assess whether data conforms to expected patterns, while profiling describes dataset structure, missingness patterns, and statistical properties. Regression analysis and hypothesis testing diagnose quality issues and quantify relationships between quality metrics and analytical outcomes.

Modern data evaluation balances automated and manual approaches. Automated evaluation offers speed, scalability, and consistency for large datasets through rule-based validation, statistical profiling, and machine learning-based anomaly detection. Manual evaluation contributes domain expertise, contextual understanding, and interpretative judgment that automated systems cannot replicate. Human-in-the-loop approaches combine automation's efficiency with human interpretability, optimizing both throughput and quality.

Tools, workflows, and governance frameworks provide infrastructure for systematic evaluation across the data lifecycle. Data profiling tools (e.g., Pandas Profiling, Great Expectations, Deequ) automate quality assessment. Validation frameworks embed checks into ETL/ELT pipelines. Data lineage tracking and metadata management support traceability and impact analysis. Governance frameworks establish roles, responsibilities, and processes aligning evaluation practices with regulatory requirements and reproducibility needs.

3 Applications in the Programming Language R

Please read the [How to Use R for Data Science](#) by Prof. Dr. Huber for any basic questions regarding R programming.

3.1 Core tidyverse Tooling

Fundamental packages:

- `dplyr` for data manipulation (filter, mutate, summarize, joins).

```
# Loading necessary library
library(dplyr)

# Using dplyr to pipe
df_summary <- df_clean %>%
  group_by(country) %>%
  summarize(
    mean_death_rate = mean(death_rate, na.rm = TRUE),
    max_death_rate = max(death_rate, na.rm = TRUE),
    min_death_rate = min(death_rate, na.rm = TRUE)
  ) %>%
  arrange(desc(mean_death_rate))

# Displaying the summary statistics
print(df_summary)
```

```
# A tibble: 46 x 4
  country          mean_death_rate max_death_rate min_death_rate
  <chr>              <dbl>          <dbl>          <dbl>
1 South Africa      330.            438            241
2 Latvia            230.            364            151
3 Lithuania         225.            340            134
4 Romania            223.            303            172
5 Hungary            218.            375            141
6 Mexico            216.            453            155
7 Bulgaria           193.            378            156
8 Brazil             185.            356            133
9 Slovak Republic   178.            308            130
10 Estonia           172.            265            100
# i 36 more rows
```

- `tidyr` for data reshaping (pivoting, nesting, separating, unnesting).

```
# Loading necessary library
library(tidyr)

# Using tidyr to pivot data
df_wide <- df_clean %>%
  pivot_wider(
```



```

    names_from = year,
    values_from = death_rate
  )

# Displaying the wide format data
print(df_wide)

# A tibble: 46 x 16
  country_code country `2010` `2011` `2012` `2013` `2014` `2015` `2016` `2017`
  <chr>         <chr>   <list> <list> <list> <list> <list> <list> <list>
1 AUS          Austral~ <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
2 AUT          Austria <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
3 BEL          Belgium <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
4 CAN          Canada  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
5 CHL          Chile   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
6 COL          Colombia <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
7 CRI          Costa R~ <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
8 CZE          Czechia <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
9 DNK          Denmark <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
10 EST         Estonia <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
# i 36 more rows
# i 6 more variables: `2018` <list>, `2019` <list>, `2020` <list>,
#   `2021` <list>, `2022` <list>, `2023` <list>

```

- `ggplot2` for layered grammar-based visualization. See the example above.
- some more R libraries:
 - `readr` for ingestion of CSV and other flat files.
 - `readxl` for ingestion of Excel files.
 - `lubridate` for date-time handling.
 - `purrr` for functional iteration.
 - `stringr` for text handling.
 - `forcats` for factor handling.

for ingestion, functional iteration, text, and factor handling.

3.2 Data Visualization Principles

Choose encodings appropriate to variable types:

- Continuous (Quantitative) variables → Position, Length Examples: x/y coordinates in scatter plots, bar heights, line positions
- Categorical (Nominal) variables → Color hue, Shape, Facets Examples: different colors for groups, point shapes, separate panels
- Ordinal variables → Ordered position, Color saturation Examples: ordered categories on axis, gradient colors from light to dark
- Temporal variables → Position along x-axis, Line connections Examples: time on horizontal axis, connected points showing progression
- Compositional (Part-to-whole) → Stacked position, Area Examples: stacked bars, proportional areas

Emphasize clarity: reduce chart junk; apply perceptual best practices:

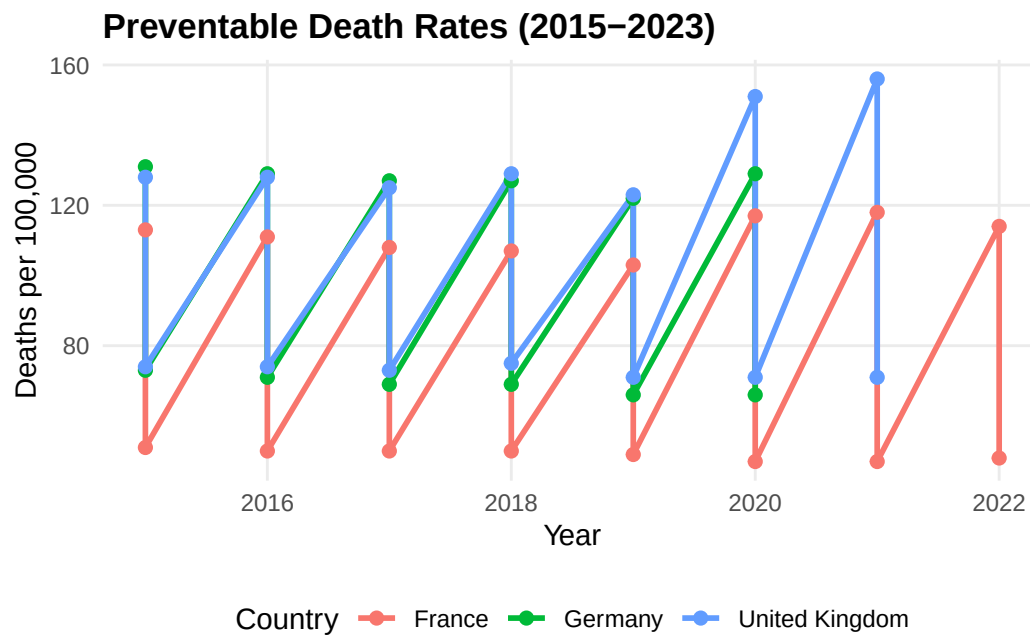
Clarity in data visualization requires removing unnecessary elements that distract from the data while applying principles of human perception to enhance understanding.

- Remove decorative elements (3D effects, shadows, gradients)
- Eliminate redundant labels and gridlines
- Minimize non-data ink (borders, background colors)
- Avoid unnecessary legends when direct labeling is possible
- Use position over angle for quantitative comparisons (bar charts > pie charts)
- Maintain consistent scales across comparable charts
- Respect aspect ratios that emphasize meaningful patterns
- Choose colorblind-friendly palettes
- Ensure sufficient contrast between data and background

```
# Loading necessary libraries
library(ggplot2)
library(dplyr)

# Creating data for demonstration
country_subset <- df_clean %>%
  filter(country %in% c("Germany", "France", "United Kingdom")) %>%
  filter(year >= 2015)

# Example: Clean, minimal visualization
ggplot(country_subset, aes(x = year, y = death_rate, color = country)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Preventable Death Rates (2015-2023)",
    x = "Year",
    y = "Deaths per 100,000",
    color = "Country"
  ) +
  theme_minimal() +
  theme(
    panel.grid.minor = element_blank(), # Removing unnecessary gridlines
    legend.position = "bottom",
    plot.title = element_text(face = "bold")
  )
```



Support comparison, trend detection, and anomaly spotting:

Effective visualizations should facilitate three key analytical tasks: comparing values across groups, identifying trends over time, and detecting unusual patterns.

Support comparison:

- Align items on common scales for direct comparison
- Use small multiples (facets) for comparing across categories
- Order categorical variables meaningfully (by value, alphabetically, or logically)
- Keep consistent ordering across related charts

Enable trend detection:

- Use connected lines for temporal data to show continuity
- Add trend lines (linear, loess) to highlight overall patterns
- Display sufficient time periods to establish meaningful trends
- Avoid over-smoothing that hides important variations

Facilitate anomaly spotting:

- Use reference lines or bands for expected ranges
- Highlight outliers through color or annotation
- Include context (confidence intervals, historical ranges)
- Maintain consistent scales to make deviations visible

```
# Loading necessary libraries
library(ggplot2)
library(dplyr)

# Calculating statistics for anomaly detection
country_stats <- df_clean %>%
  group_by(country) %>%
  summarise(
```

```

    mean_rate = mean(death_rate, na.rm = TRUE),
    sd_rate = sd(death_rate, na.rm = TRUE)
  )

# Joining back to identify anomalies
df_annotated <- df_clean %>%
  left_join(country_stats, by = "country") %>%
  mutate(
    z_score = (death_rate - mean_rate) / sd_rate,
    is_anomaly = abs(z_score) > 2 # Flagging values > 2 standard deviations
  )

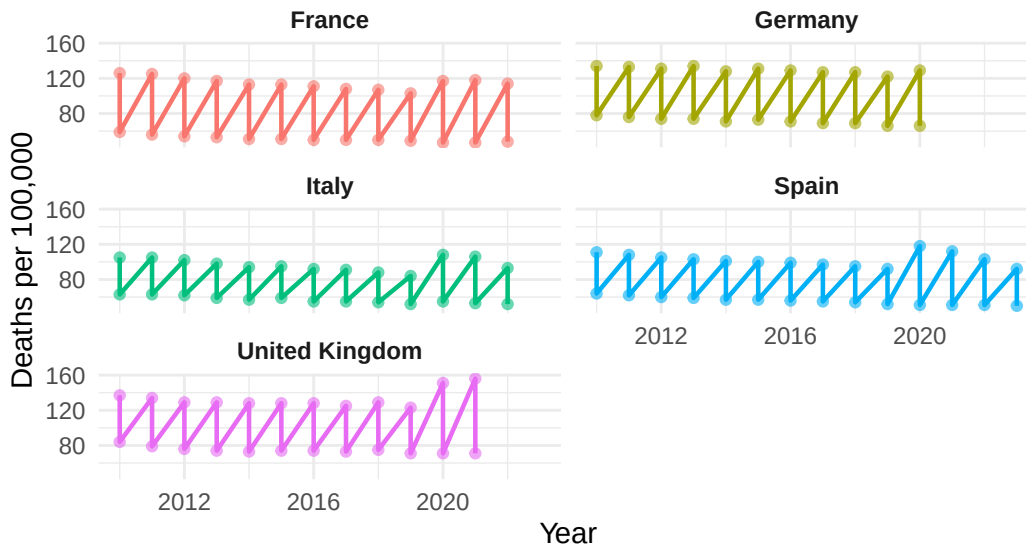
# Example: Visualization supporting comparison, trends, and anomaly detection
selected_countries <- c("Germany", "France", "United Kingdom", "Italy", "Spain")
df_subset <- df_annotated %>% filter(country %in% selected_countries)

ggplot(df_subset, aes(x = year, y = death_rate, color = country)) +
  geom_line(linewidth = 0.8) +
  geom_point(aes(size = is_anomaly, alpha = is_anomaly)) +
  scale_size_manual(values = c(1.5, 3), guide = "none") +
  scale_alpha_manual(values = c(0.6, 1), guide = "none") +
  facet_wrap(~ country, ncol = 2) + # Small multiples for comparison
  labs(
    title = "Preventable Death Rates: Trends and Anomalies",
    subtitle = "Larger points indicate statistical anomalies (>2 SD from country mean)",
    x = "Year",
    y = "Deaths per 100,000",
    color = "Country"
  ) +
  theme_minimal() +
  theme(
    legend.position = "none",
    strip.text = element_text(face = "bold"),
    plot.title = element_text(face = "bold")
  )

```

Preventable Death Rates: Trends and Anomalies

Larger points indicate statistical anomalies (>2 SD from country mean)



3.3 Detecting Outliers and Anomalies

- **Rule-based methods (IQR, z-scores):** These classical univariate approaches identify outliers by establishing statistical thresholds, where observations falling beyond predefined boundaries (e.g., $1.5 \times \text{IQR}$ or ± 3 standard deviations) are flagged as potential anomalies. While computationally efficient and interpretable, these methods assume underlying distributional properties and may overlook multivariate patterns.

```
# Loading necessary library
library(dplyr)

# Calculating z-scores for death_rate
df_zscores <- df_clean %>%
  group_by(country) %>%
  mutate(
    mean_rate = mean(death_rate, na.rm = TRUE),
    sd_rate = sd(death_rate, na.rm = TRUE),
    z_score = (death_rate - mean_rate) / sd_rate,
    is_outlier = abs(z_score) > 3 # Flagging z-scores beyond ±3
  ) %>%
  ungroup()

# Displaying observations flagged as outliers
outliers <- df_zscores %>%
  filter(is_outlier)
print(outliers)
```

A tibble: 10 x 8

country_code	country	year	death_rate	mean_rate	sd_rate	z_score	is_outlier
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<lgl>

1	COL	Colombia	2021	304	142.	48.2	3.35	TRUE
2	CRI	Costa Rica	2021	209	114.	28.3	3.36	TRUE
3	MEX	Mexico	2020	445	216.	75.5	3.03	TRUE
4	MEX	Mexico	2021	453	216.	75.5	3.14	TRUE
5	SVK	Slovak Re~	2021	308	178.	43.4	3.00	TRUE
6	ARG	Argentina	2021	269	149.	27.1	4.42	TRUE
7	BRA	Brazil	2021	356	185.	51.6	3.31	TRUE
8	BGR	Bulgaria	2021	378	193.	45.5	4.07	TRUE
9	PER	Peru	2020	408	124.	90.4	3.14	TRUE
10	PER	Peru	2021	447	124.	90.4	3.57	TRUE

- **Robust statistics (median, MAD):** Resistant measures such as the median and median absolute deviation (MAD) provide reliable central tendency and dispersion estimates that are less influenced by extreme values compared to mean and standard deviation. These statistics form the foundation for outlier detection in skewed or heavy-tailed distributions where parametric assumptions are violated.

```
# Loading necessary library
library(dplyr)

# Calculating robust statistics for death_rate
df_robust <- df_clean %>%
  group_by(country) %>%
  mutate(
    median_rate = median(death_rate, na.rm = TRUE),
    mad_rate = mad(death_rate, constant = 1, na.rm = TRUE),
    robust_z = 0.6745 * (death_rate - median_rate) / mad_rate,
    is_robust_outlier = abs(robust_z) > 3.5
  ) %>%
  ungroup()

# Displaying observations flagged as robust outliers
robust_outliers <- df_robust %>%
  filter(is_robust_outlier)
print(robust_outliers)
```

```
# A tibble: 10 x 8
  country_code country    year death_rate median_rate mad_rate robust_z
  <chr>         <chr>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
1 COL          Colombia  2021      304      132      27      4.30
2 MEX          Mexico    2020      445      217      34      4.52
3 MEX          Mexico    2021      453      217      34      4.68
4 ARG          Argentina 2020      194      144.      5.5      6.19
5 ARG          Argentina 2021      269      144.      5.5     15.4
6 BRA          Brazil    2021      356      172.      31      4.01
7 BGR          Bulgaria  2021      378      186      20.5     6.32
8 PER          Peru      2020      408      97.5      6.5     32.2
9 PER          Peru      2021      447      97.5      6.5     36.3
10 PER         Peru      2022      146      97.5      6.5      5.03

# i 1 more variable: is_robust_outlier <lgl>
```

- **Model-based or multivariate detection (e.g., Mahalanobis distance, clustering residuals):** Advanced techniques account for correlation structures and multidimensional relationships, enabling detection of outliers that appear normal in individual dimensions but are anomalous in multivariate space. Mahalanobis distance measures how many standard deviations an observation is from the distribution center, while clustering residuals identify observations that deviate from expected cluster membership patterns.
- **Distinguish errors vs. novel but valid observations:** Critical analytical judgment is required to differentiate between measurement errors, data entry mistakes, and legitimate extreme values that represent rare but genuine phenomena. This distinction has profound implications for data quality management and scientific discovery, as premature removal of valid outliers may obscure important patterns or emerging trends.

3.4 Dimensionality Reduction

- **Motivation: mitigate multicollinearity, noise, and curse of dimensionality:** High-dimensional datasets present computational challenges and statistical complications, including increased sparsity, model overfitting, and reduced discriminatory power of distance-based methods. Dimensionality reduction addresses these issues by transforming data into lower-dimensional representations while preserving essential variance and structural relationships.
- **Techniques: Principal Component Analysis (PCA), Factor Analysis, (optionally) t-SNE / UMAP (for exploration):** PCA identifies orthogonal linear combinations of variables that maximize variance, creating uncorrelated components suitable for regression and classification tasks. Factor Analysis assumes latent constructs underlying observed variables, focusing on shared variance and theoretical interpretation, while non-linear methods like t-SNE and UMAP preserve local neighborhood structures for exploratory visualization of complex data manifolds.

```
# Loading necessary libraries
library(dplyr)

# Preparing data for PCA (using numeric variables only)
df_pca <- df_clean %>%
  select(year, death_rate) %>%
  na.omit()

# Performing PCA
pca_result <- prcomp(df_pca, scale. = TRUE)

# Displaying summary of PCA
summary(pca_result)
```

Importance of components:

	PC1	PC2
Standard deviation	1.0189	0.9807
Proportion of Variance	0.5191	0.4809
Cumulative Proportion	0.5191	1.0000

```
# Showing principal component loadings
print(pca_result$rotation)
```

```

              PC1      PC2
year          0.7071068 0.7071068
death_rate -0.7071068 0.7071068

```

```
# Calculating proportion of variance explained
var_explained <- pca_result$sdev^2 / sum(pca_result$sdev^2)
cat("Proportion of variance explained by PC1:", round(var_explained[1], 3), "\n")
```

Proportion of variance explained by PC1: 0.519

```
cat("Proportion of variance explained by PC2:", round(var_explained[2], 3), "\n")
```

Proportion of variance explained by PC2: 0.481

- **Interpretability vs. compression trade-offs:** Dimensionality reduction inherently balances the competing objectives of achieving parsimonious data representations and maintaining interpretable relationships to original variables. While aggressive compression maximizes computational efficiency and reduces noise, it may obscure meaningful features and complicate domain-specific interpretation of analytical results.

3.5 Data Exploration and Mining

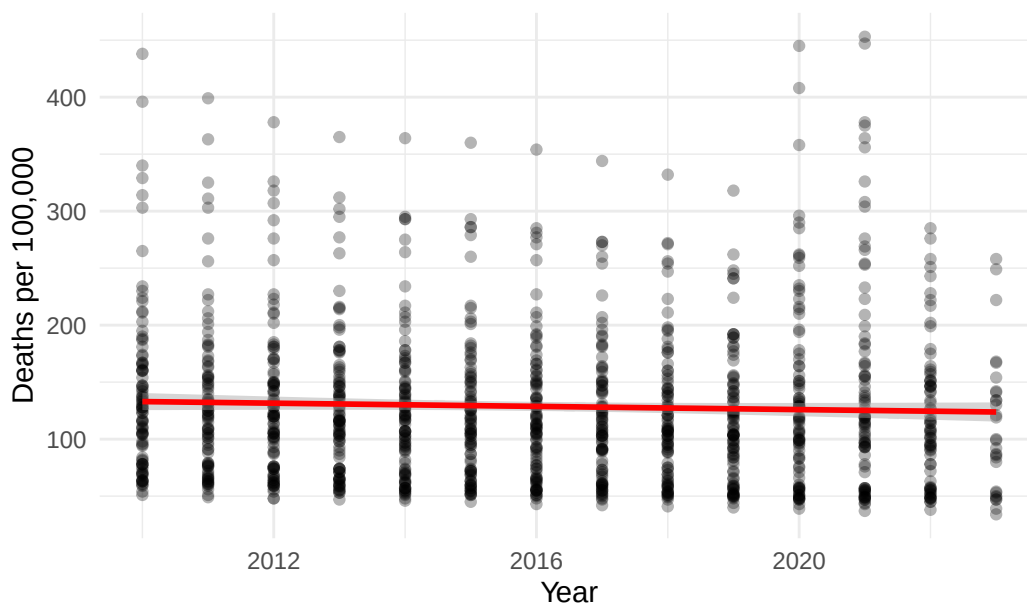
- **Structured EDA workflow: question → visualize → quantify → refine:** Exploratory Data Analysis follows a systematic iterative process that begins with research questions, employs visualization to generate hypotheses, quantifies patterns through statistical measures, and refines understanding through successive analytical cycles. This disciplined approach prevents data dredging while ensuring comprehensive investigation of data characteristics and relationships.

```
# Loading necessary libraries
library(ggplot2)
library(dplyr)

# Step 1: Question - Are death rates declining over time?

# Step 2: Visualize
ggplot(df_clean, aes(x = year, y = death_rate)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "EDA: Death Rates Over Time",
    x = "Year",
    y = "Deaths per 100,000"
  ) +
  theme_minimal()
```


EDA: Death Rates Over Time



```
# Step 3: Quantify
trend_model <- lm(death_rate ~ year, data = df_clean)
summary(trend_model)
```

Call:

```
lm(formula = death_rate ~ year, data = df_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-89.80	-54.93	-13.16	27.95	327.80

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1544.5523	1087.6865	1.420	0.156
year	-0.7023	0.5395	-1.302	0.193

Residual standard error: 69.57 on 1160 degrees of freedom

Multiple R-squared: 0.001459, Adjusted R-squared: 0.0005978

F-statistic: 1.694 on 1 and 1160 DF, p-value: 0.1933

```
# Step 4: Refine - Examine by country
country_trends <- df_clean %>%
  group_by(country) %>%
  summarise(
    correlation = cor(year, death_rate, use = "complete.obs")
  ) %>%
  arrange(correlation)

print(head(country_trends))
```

```
# A tibble: 6 x 2
  country      correlation
  <chr>        <dbl>
1 South Africa -0.663
2 Israel       -0.473
3 Korea        -0.338
4 Japan        -0.324
5 Luxembourg   -0.317
6 Lithuania    -0.290
```

- **PCA for variance structure:** Principal Component Analysis reveals the underlying variance-covariance structure of multivariate data, identifying dimensions of maximum variability and reducing redundancy among correlated variables. By examining eigenvalues and component loadings, analysts determine the effective dimensionality of the dataset and detect dominant patterns in the data.
- **Factor Analysis for latent constructs:** This technique assumes that observed variables are manifestations of unobservable latent factors, making it particularly valuable for psychometric research and construct validation. Unlike PCA, Factor Analysis models measurement error explicitly and focuses on shared rather than total variance, facilitating theoretical interpretation of underlying psychological or economic constructs.

```
# Loading necessary libraries
library(dplyr)
library(tidyr)

# Creating a wider dataset for factor analysis
# First, get unique country-year combinations by averaging any duplicates
df_wide_fa <- df_clean %>%
  filter(year >= 2018) %>%
  select(country, year, death_rate) %>%
  group_by(country, year) %>%
  summarise(death_rate = mean(death_rate, na.rm = TRUE), .groups = "drop") %>%
  pivot_wider(names_from = year, values_from = death_rate, names_prefix = "year_") %>%
  select(-country) %>%
  na.omit()

# Performing factor analysis using base R stats package
if(nrow(df_wide_fa) > 10 && ncol(df_wide_fa) >= 3) {
  # Determining number of factors (minimum of 2 or ncol-1)
  n_factors <- min(2, ncol(df_wide_fa) - 1)

  # Using factanal from stats package
  fa_result <- factanal(df_wide_fa, factors = n_factors, rotation = "varimax")

  # Displaying factor loadings
  cat("Factor Loadings:\n")
  print(fa_result$loadings)

  # Calculating proportion of variance explained
  loadings_sq <- fa_result$loadings^2
  var_explained <- colSums(loadings_sq) / nrow(loadings_sq)
```

```

cat("\nProportion of variance explained by each factor:\n")
print(var_explained)

# Displaying uniquenesses (proportion of variance not explained by factors)
cat("\nUniqueness (1 - communality):\n")
print(fa_result$uniquenesses)
} else {
  cat("Note: Not enough observations or variables for factor analysis.\n")
  cat("Factor analysis requires at least 3 variables and sufficient observations.\n")
}

```

Factor Loadings:

Loadings:

	Factor1	Factor2
year_2018	0.759	0.650
year_2019	0.754	0.655
year_2020	0.673	0.736
year_2021	0.731	0.668
year_2022	0.775	0.630
year_2023	0.729	0.664

	Factor1	Factor2
SS loadings	3.262	2.676
Proportion Var	0.544	0.446
Cumulative Var	0.544	0.990

Proportion of variance explained by each factor:

Factor1	Factor2
0.5437206	0.4460732

Uniqueness (1 - communality):

year_2018	year_2019	year_2020	year_2021	year_2022	year_2023
0.00500000	0.00500000	0.00500000	0.01875942	0.00500000	0.02905113

- **Regression Analysis for relationships and predictive structure:** Linear and non-linear regression models quantify relationships between dependent and independent variables, enabling both explanatory analysis of associations and predictive modeling of outcomes. These methods provide parameter estimates, statistical inference, and diagnostic tools to assess model adequacy and identify influential observations.

```

# Loading necessary library
library(dplyr)

# Preparing data with a categorical variable
df_regression <- df_clean %>%
  mutate(
    time_period = case_when(
      year < 2015 ~ "Early",
      year >= 2015 & year < 2020 ~ "Middle",

```

```

    year >= 2020 ~ "Recent"
  )
) %>%
filter(!is.na(time_period))

# Simple linear regression
model_simple <- lm(death_rate ~ year, data = df_regression)
summary(model_simple)

```

Call:

```
lm(formula = death_rate ~ year, data = df_regression)
```

Residuals:

Min	1Q	Median	3Q	Max
-89.80	-54.93	-13.16	27.95	327.80

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1544.5523	1087.6865	1.420	0.156
year	-0.7023	0.5395	-1.302	0.193

Residual standard error: 69.57 on 1160 degrees of freedom

Multiple R-squared: 0.001459, Adjusted R-squared: 0.0005978

F-statistic: 1.694 on 1 and 1160 DF, p-value: 0.1933

```

# Multiple regression with categorical predictor
model_multiple <- lm(death_rate ~ year + time_period, data = df_regression)
summary(model_multiple)

```

Call:

```
lm(formula = death_rate ~ year + time_period, data = df_regression)
```

Residuals:

Min	1Q	Median	3Q	Max
-98.10	-54.76	-12.67	27.19	319.30

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6960.638	3073.466	2.265	0.0237 *
year	-3.393	1.528	-2.221	0.0265 *
time_periodMiddle	5.755	8.892	0.647	0.5177
time_periodRecent	31.218	14.975	2.085	0.0373 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 69.32 on 1158 degrees of freedom

Multiple R-squared: 0.01036, Adjusted R-squared: 0.007793

F-statistic: 4.04 on 3 and 1158 DF, p-value: 0.007176

```
# Extracting and interpreting coefficients
cat("\nCoefficient interpretation:\n")
```

Coefficient interpretation:

```
cat("Year coefficient:", round(coef(model_simple)[2], 3), "\n")
```

Year coefficient: -0.702

```
cat("This means death rate changes by", round(coef(model_simple)[2], 3),
    "per 100,000 per year\n")
```

This means death rate changes by -0.702 per 100,000 per year

- **Clustering (k-means, hierarchical) for pattern discovery (if included):** Unsupervised clustering algorithms partition observations into homogeneous groups based on similarity metrics, revealing natural taxonomies and segment structures within data. K-means optimizes within-cluster variance through iterative assignment, while hierarchical methods create nested groupings that can be visualized through dendrograms to inform cluster selection.

```
# Loading necessary libraries
library(dplyr)
library(ggplot2)

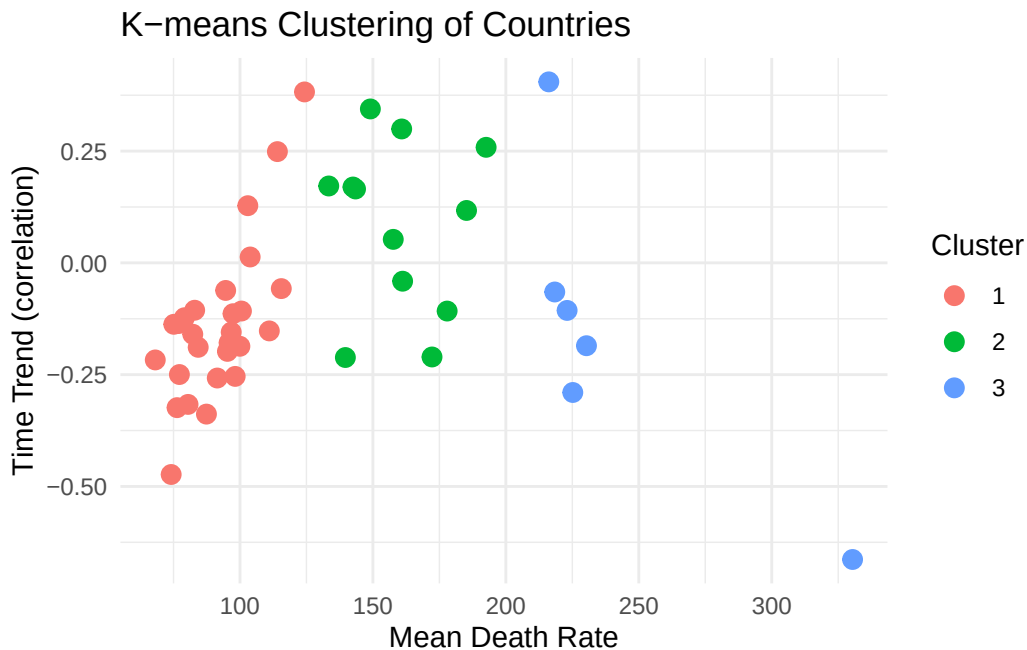
# Preparing data for clustering - average death rate by country
df_cluster <- df_clean %>%
  group_by(country) %>%
  summarise(
    mean_death_rate = mean(death_rate, na.rm = TRUE),
    trend = cor(year, death_rate, use = "complete.obs")
  ) %>%
  na.omit()

# K-means clustering
set.seed(123)
kmeans_result <- kmeans(df_cluster[, c("mean_death_rate", "trend")], centers = 3)

# Adding cluster assignment to data
df_cluster$cluster <- as.factor(kmeans_result$cluster)

# Visualizing clusters
ggplot(df_cluster, aes(x = mean_death_rate, y = trend, color = cluster)) +
  geom_point(size = 3) +
  labs(
    title = "K-means Clustering of Countries",
    x = "Mean Death Rate",
    y = "Time Trend (correlation)",
```

```
color = "Cluster"
) +
theme_minimal()
```



```
# Displaying cluster centers
cat("\nCluster Centers:\n")
```

Cluster Centers:

```
print(kmeans_result$centers)
```

	mean_death_rate	trend
1	91.57995	-0.13751640
2	159.63393	0.08403368
3	240.58291	-0.15086402

3.6 Causal Inference with Regression Analysis

- **Distinguish association vs. causation:** Statistical association measures correlation between variables without implying directionality, whereas causal inference attempts to establish that changes in one variable directly produce changes in another. Demonstrating causality requires careful consideration of temporal precedence, theoretical mechanisms, and elimination of alternative explanations through research design and statistical controls.

```
# Loading necessary library
library(dplyr)

# Example: Association between year and death rate
association <- cor(df_clean$year, df_clean$death_rate, use = "complete.obs")
cat("Association (correlation) between year and death rate:", round(association, 3), "\n")
```

Association (correlation) between year and death rate: -0.038

```
cat("\nThis shows association, but does NOT prove that time causes death rate changes.\n")
```

This shows association, but does NOT prove that time causes death rate changes.

```
cat("Potential confounders: healthcare improvements, policy changes, etc.\n")
```

Potential confounders: healthcare improvements, policy changes, etc.

```
# Demonstrating how confounders can affect interpretation
# Creating a hypothetical scenario
df_confound <- df_clean %>%
  mutate(
    developed = country %in% c("Germany", "France", "United Kingdom",
                              "United States", "Japan")
  )

# Correlation in developed vs developing countries
cor_developed <- df_confound %>%
  filter(developed == TRUE) %>%
  summarise(cor = cor(year, death_rate, use = "complete.obs")) %>%
  pull(cor)

cor_developing <- df_confound %>%
  filter(developed == FALSE) %>%
  summarise(cor = cor(year, death_rate, use = "complete.obs")) %>%
  pull(cor)

cat("\nCorrelation in developed countries:", round(cor_developed, 3), "\n")
```

Correlation in developed countries: 0.002

```
cat("Correlation in developing countries:", round(cor_developing, 3), "\n")
```

Correlation in developing countries: -0.046

```
cat("\nDifferent correlations suggest development level may be a confounder.\n")
```

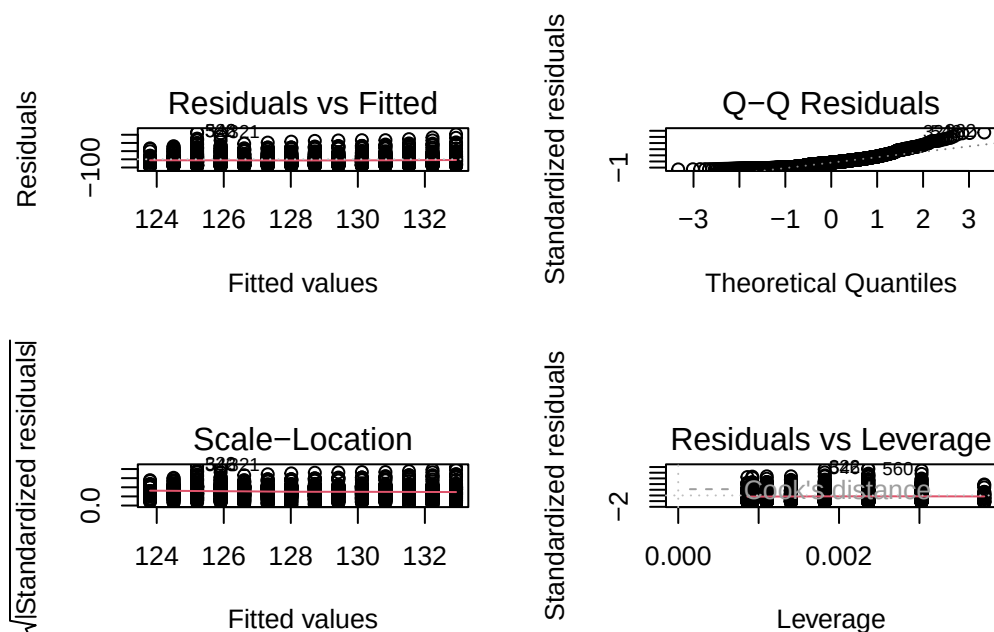
Different correlations suggest development level may be a confounder.

- **Model specification and confounding control:** Proper model specification identifies relevant covariates and functional forms to isolate the causal effect of interest while controlling for confounding variables that influence both treatment and outcome. Omitted variable bias, measurement error, and incorrect functional forms threaten causal identification, necessitating theory-driven variable selection and specification testing.
- **Assumptions: linearity, independence, homoskedasticity, exogeneity:** Valid causal inference from regression requires that relationships are linear in parameters, observations are independent, error variance is constant across predictor levels, and explanatory variables are uncorrelated with the error term. Violations of these assumptions bias coefficient estimates, invalidate standard errors, and compromise hypothesis tests, requiring diagnostic assessment and remedial measures.

```
# Loading necessary libraries
library(ggplot2)
library(dplyr)
library(lmtest)

# Fitting a regression model
model <- lm(death_rate ~ year, data = df_clean)

# Checking assumptions through diagnostic plots
par(mfrow = c(2, 2))
plot(model)
```




```
par(mfrow = c(1, 1))

# Testing for homoskedasticity (Breusch-Pagan test)
bp_test <- bptest(model)
cat("\nBreusch-Pagan test for homoskedasticity:\n")
```

Breusch-Pagan test for homoskedasticity:

```
cat("p-value:", bp_test$p.value, "\n")
```

p-value: 0.2263175

```
if(bp_test$p.value < 0.05) {
  cat("Evidence of heteroskedasticity (non-constant variance)\n")
} else {
  cat("No strong evidence against homoskedasticity\n")
}
```

No strong evidence against homoskedasticity

```
# Checking for normality of residuals
shapiro_test <- shapiro.test(residuals(model)[1:5000]) # Shapiro test limited to 5000 obs
cat("\nShapiro-Wilk test for normality of residuals:\n")
```

Shapiro-Wilk test for normality of residuals:

```
cat("p-value:", shapiro_test$p.value, "\n")
```

p-value: 9.063082e-29

- **Interpretation of coefficients and marginal effects:** Regression coefficients represent the expected change in the dependent variable associated with a one-unit change in the independent variable, holding other factors constant. Marginal effects extend this interpretation to non-linear models and interaction terms, quantifying how the impact of one variable varies across levels of another variable.

```
# Loading necessary library
library(dplyr)

# Creating interaction term
df_interaction <- df_clean %>%
  mutate(
    recent_period = ifelse(year >= 2020, 1, 0),
    year_centered = year - mean(year)
  )
```

```
# Model with interaction
model_interaction <- lm(death_rate ~ year_centered * recent_period,
                        data = df_interaction)
summary(model_interaction)
```

Call:

```
lm(formula = death_rate ~ year_centered * recent_period, data = df_interaction)
```

Residuals:

Min	1Q	Median	3Q	Max
-104.90	-54.62	-13.11	28.45	318.36

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	123.948	2.607	47.541	< 2e-16 ***
year_centered	-2.313	0.807	-2.866	0.00423 **
recent_period	56.979	22.746	2.505	0.01238 *
year_centered:recent_period	-6.947	4.376	-1.588	0.11267

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 69.25 on 1158 degrees of freedom

Multiple R-squared: 0.01215, Adjusted R-squared: 0.00959

F-statistic: 4.747 on 3 and 1158 DF, p-value: 0.002694

```
# Interpreting coefficients
coefs <- coef(model_interaction)
cat("\nCoefficient Interpretation:\n")
```

Coefficient Interpretation:

```
cat("-----\n")
```

```
cat("Main effect of year:", round(coefs["year_centered"], 3), "\n")
```

Main effect of year: -2.313

```
cat(" -> In pre-2020 period, each year is associated with a",
     round(coefs["year_centered"], 3), "change in death rate\n\n")
```

-> In pre-2020 period, each year is associated with a -2.313 change in death rate

```
cat("Interaction effect:", round(coefs["year_centered:recent_period"], 3), "\n")
```

Interaction effect: -6.947

```
cat("  -> In 2020+, the year effect changes by an additional",  
    round(coefs["year_centered:recent_period"], 3), "\n")
```

-> In 2020+, the year effect changes by an additional -6.947

```
cat("  -> Total effect in 2020+:",  
    round(coefs["year_centered"] + coefs["year_centered:recent_period"], 3), "\n")
```

-> Total effect in 2020+: -9.26

- **Sensitivity and robustness checks:** Assessing the stability of causal conclusions across alternative model specifications, sample restrictions, and analytical choices strengthens inference and identifies fragile results dependent on specific assumptions. Techniques include varying control variables, testing different functional forms, examining subgroup effects, and conducting placebo tests to validate identification strategies.

```
# Loading necessary library  
library(dplyr)  
  
# Base model  
model1 <- lm(death_rate ~ year, data = df_clean)  
  
# Alternative specification 1: Adding squared term  
df_robust <- df_clean %>%  
  mutate(year_sq = year^2)  
model2 <- lm(death_rate ~ year + year_sq, data = df_robust)  
  
# Alternative specification 2: Different time periods  
df_subset1 <- df_clean %>% filter(year >= 2015)  
model3 <- lm(death_rate ~ year, data = df_subset1)  
  
df_subset2 <- df_clean %>% filter(year < 2020)  
model4 <- lm(death_rate ~ year, data = df_subset2)  
  
# Comparing coefficients across models  
cat("Robustness Check: Year Coefficient Across Specifications\n")
```

Robustness Check: Year Coefficient Across Specifications

```
cat("-----\n")
```

```
cat("Model 1 (Base):", round(coef(model1)["year"], 4), "\n")
```

Model 1 (Base): -0.7023

```
cat("Model 2 (Quadratic):", round(coef(model2)["year"], 4), "\n")
```

Model 2 (Quadratic): -1105.846

```
cat("Model 3 (2015+):", round(coef(model3)["year"], 4), "\n")
```

Model 3 (2015+): 0.9826

```
cat("Model 4 (Pre-2020):", round(coef(model4)["year"], 4), "\n")
```

Model 4 (Pre-2020): -2.3126

```
cat("\nInterpretation: If coefficients are similar across specifications,\n")
```

Interpretation: If coefficients are similar across specifications,

```
cat("this suggests the relationship is robust to model choices.\n")
```

this suggests the relationship is robust to model choices.

```
# Comparing R-squared values  
cat("\nModel Fit Comparison:\n")
```

Model Fit Comparison:

```
cat("Model 1 R-squared:", round(summary(model1)$r.squared, 4), "\n")
```

Model 1 R-squared: 0.0015

```
cat("Model 2 R-squared:", round(summary(model2)$r.squared, 4), "\n")
```

Model 2 R-squared: 0.0042

4 Project report and presentation

You will find all formal instructions for the project report and presentation in the [guidelines](#).

Presentation timetable:

- Project *Spotify* (Jill, Leonie, Isabel): Week 49 (3.12.2025)
- Project *Environmental Pollution* (Frederik): Week 49 (3.12.2025)
- Project *Elon Musk* (Esra, Mira): Week 50 (10.12.2025)
- Project *Influencer Engagement* (Stella, Athina): Week 50 (10.12.2025)
- Project *Soccer Player Valuation* (Dominic): Week 51 (10.12.2025)
- Project *COVID-19 pandemic* (Mats, Laurine, Eric): Week 51 (17.12.2025)
- Project *AI companion* (Jamder): Week 51 (17.12.2025)

Assessment criteria for the report:

- Clarity of Research Question (and Objectives)
- Data Understanding and Preparation
- Appropriateness of Methods and Analysis
- Implementation and Code Quality (in R)
- Interpretation of Results
- Visualization and Communication
- Structure and Presentation of the Report
- Critical Reflection and Limitations

Example linear regression analysis and storytelling:

```
# Loading necessary libraries
library(ourdata)
library(dplyr)

# Opening help of package
??ourdata
oecd_preventable
```

	STRUCTURE	STRUCTURE_ID	STRUCTURE_NAME
1	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
2	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
3	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
4	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
5	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
6	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
7	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
8	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
9	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
10	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
11	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
12	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
13	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
14	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
15	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	
16	DATAFLOW OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality	

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1139	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1140	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1141	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1142	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1143	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1144	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1145	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1146	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1147	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1148	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1149	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1150	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1151	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1152	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1153	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1154	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1155	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1156	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1157	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1158	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1159	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1160	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1161	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
1162	DATAFLOW	OECD.ELS.HD:DSD_HEALTH_STAT@DF_AM(1.1)	Avoidable mortality
ACTION REF_AREA Reference.area FREQ Frequency.of.observation MEASURE			
1	I	AUS	Australia A Annual PREVM
2	I	AUS	Australia A Annual PREVM
3	I	AUS	Australia A Annual PREVM
4	I	AUS	Australia A Annual PREVM
5	I	AUS	Australia A Annual PREVM
6	I	AUS	Australia A Annual PREVM
7	I	AUS	Australia A Annual PREVM
8	I	AUS	Australia A Annual PREVM
9	I	AUS	Australia A Annual PREVM
10	I	AUS	Australia A Annual PREVM
11	I	AUS	Australia A Annual PREVM
12	I	AUS	Australia A Annual PREVM
13	I	AUS	Australia A Annual PREVM
14	I	AUS	Australia A Annual PREVM
15	I	AUT	Austria A Annual PREVM
16	I	AUT	Austria A Annual PREVM
17	I	AUT	Austria A Annual PREVM
18	I	AUT	Austria A Annual PREVM
19	I	AUT	Austria A Annual PREVM
20	I	AUT	Austria A Annual PREVM
21	I	AUT	Austria A Annual PREVM
22	I	AUT	Austria A Annual PREVM
23	I	AUT	Austria A Annual PREVM
24	I	AUT	Austria A Annual PREVM
25	I	AUT	Austria A Annual PREVM
26	I	AUT	Austria A Annual PREVM

27	I	AUT	Austria	A	Annual	PREVM
28	I	AUT	Austria	A	Annual	PREVM
29	I	BEL	Belgium	A	Annual	PREVM
30	I	BEL	Belgium	A	Annual	PREVM
31	I	BEL	Belgium	A	Annual	PREVM
32	I	BEL	Belgium	A	Annual	PREVM
33	I	BEL	Belgium	A	Annual	PREVM
34	I	BEL	Belgium	A	Annual	PREVM
35	I	BEL	Belgium	A	Annual	PREVM
36	I	BEL	Belgium	A	Annual	PREVM
37	I	BEL	Belgium	A	Annual	PREVM
38	I	BEL	Belgium	A	Annual	PREVM
39	I	BEL	Belgium	A	Annual	PREVM
40	I	BEL	Belgium	A	Annual	PREVM
41	I	CAN	Canada	A	Annual	PREVM
42	I	CAN	Canada	A	Annual	PREVM
43	I	CAN	Canada	A	Annual	PREVM
44	I	CAN	Canada	A	Annual	PREVM
45	I	CAN	Canada	A	Annual	PREVM
46	I	CAN	Canada	A	Annual	PREVM
47	I	CAN	Canada	A	Annual	PREVM
48	I	CAN	Canada	A	Annual	PREVM
49	I	CAN	Canada	A	Annual	PREVM
50	I	CAN	Canada	A	Annual	PREVM
51	I	CAN	Canada	A	Annual	PREVM
52	I	CAN	Canada	A	Annual	PREVM
53	I	CAN	Canada	A	Annual	PREVM
54	I	CHL	Chile	A	Annual	PREVM
55	I	CHL	Chile	A	Annual	PREVM
56	I	CHL	Chile	A	Annual	PREVM
57	I	CHL	Chile	A	Annual	PREVM
58	I	CHL	Chile	A	Annual	PREVM
59	I	CHL	Chile	A	Annual	PREVM
60	I	CHL	Chile	A	Annual	PREVM
61	I	CHL	Chile	A	Annual	PREVM
62	I	CHL	Chile	A	Annual	PREVM
63	I	CHL	Chile	A	Annual	PREVM
64	I	CHL	Chile	A	Annual	PREVM
65	I	CHL	Chile	A	Annual	PREVM
66	I	CHL	Chile	A	Annual	PREVM
67	I	COL	Colombia	A	Annual	PREVM
68	I	COL	Colombia	A	Annual	PREVM
69	I	COL	Colombia	A	Annual	PREVM
70	I	COL	Colombia	A	Annual	PREVM
71	I	COL	Colombia	A	Annual	PREVM
72	I	COL	Colombia	A	Annual	PREVM
73	I	COL	Colombia	A	Annual	PREVM
74	I	COL	Colombia	A	Annual	PREVM
75	I	COL	Colombia	A	Annual	PREVM
76	I	COL	Colombia	A	Annual	PREVM
77	I	COL	Colombia	A	Annual	PREVM

78	I	COL	Colombia	A	Annual	PREVM
79	I	CRI	Costa Rica	A	Annual	PREVM
80	I	CRI	Costa Rica	A	Annual	PREVM
81	I	CRI	Costa Rica	A	Annual	PREVM
82	I	CRI	Costa Rica	A	Annual	PREVM
83	I	CRI	Costa Rica	A	Annual	PREVM
84	I	CRI	Costa Rica	A	Annual	PREVM
85	I	CRI	Costa Rica	A	Annual	PREVM
86	I	CRI	Costa Rica	A	Annual	PREVM
87	I	CRI	Costa Rica	A	Annual	PREVM
88	I	CRI	Costa Rica	A	Annual	PREVM
89	I	CRI	Costa Rica	A	Annual	PREVM
90	I	CRI	Costa Rica	A	Annual	PREVM
91	I	CRI	Costa Rica	A	Annual	PREVM
92	I	CZE	Czechia	A	Annual	PREVM
93	I	CZE	Czechia	A	Annual	PREVM
94	I	CZE	Czechia	A	Annual	PREVM
95	I	CZE	Czechia	A	Annual	PREVM
96	I	CZE	Czechia	A	Annual	PREVM
97	I	CZE	Czechia	A	Annual	PREVM
98	I	CZE	Czechia	A	Annual	PREVM
99	I	CZE	Czechia	A	Annual	PREVM
100	I	CZE	Czechia	A	Annual	PREVM
101	I	CZE	Czechia	A	Annual	PREVM
102	I	CZE	Czechia	A	Annual	PREVM
103	I	CZE	Czechia	A	Annual	PREVM
104	I	CZE	Czechia	A	Annual	PREVM
105	I	CZE	Czechia	A	Annual	PREVM
106	I	DNK	Denmark	A	Annual	PREVM
107	I	DNK	Denmark	A	Annual	PREVM
108	I	DNK	Denmark	A	Annual	PREVM
109	I	DNK	Denmark	A	Annual	PREVM
110	I	DNK	Denmark	A	Annual	PREVM
111	I	DNK	Denmark	A	Annual	PREVM
112	I	DNK	Denmark	A	Annual	PREVM
113	I	DNK	Denmark	A	Annual	PREVM
114	I	DNK	Denmark	A	Annual	PREVM
115	I	DNK	Denmark	A	Annual	PREVM
116	I	DNK	Denmark	A	Annual	PREVM
117	I	DNK	Denmark	A	Annual	PREVM
118	I	DNK	Denmark	A	Annual	PREVM
119	I	EST	Estonia	A	Annual	PREVM
120	I	EST	Estonia	A	Annual	PREVM
121	I	EST	Estonia	A	Annual	PREVM
122	I	EST	Estonia	A	Annual	PREVM
123	I	EST	Estonia	A	Annual	PREVM
124	I	EST	Estonia	A	Annual	PREVM
125	I	EST	Estonia	A	Annual	PREVM
126	I	EST	Estonia	A	Annual	PREVM
127	I	EST	Estonia	A	Annual	PREVM
128	I	EST	Estonia	A	Annual	PREVM

129	I	EST	Estonia	A	Annual	PREVM
130	I	EST	Estonia	A	Annual	PREVM
131	I	EST	Estonia	A	Annual	PREVM
132	I	FIN	Finland	A	Annual	PREVM
133	I	FIN	Finland	A	Annual	PREVM
134	I	FIN	Finland	A	Annual	PREVM
135	I	FIN	Finland	A	Annual	PREVM
136	I	FIN	Finland	A	Annual	PREVM
137	I	FIN	Finland	A	Annual	PREVM
138	I	FIN	Finland	A	Annual	PREVM
139	I	FIN	Finland	A	Annual	PREVM
140	I	FIN	Finland	A	Annual	PREVM
141	I	FIN	Finland	A	Annual	PREVM
142	I	FIN	Finland	A	Annual	PREVM
143	I	FIN	Finland	A	Annual	PREVM
144	I	FIN	Finland	A	Annual	PREVM
145	I	FRA	France	A	Annual	PREVM
146	I	FRA	France	A	Annual	PREVM
147	I	FRA	France	A	Annual	PREVM
148	I	FRA	France	A	Annual	PREVM
149	I	FRA	France	A	Annual	PREVM
150	I	FRA	France	A	Annual	PREVM
151	I	FRA	France	A	Annual	PREVM
152	I	FRA	France	A	Annual	PREVM
153	I	FRA	France	A	Annual	PREVM
154	I	FRA	France	A	Annual	PREVM
155	I	FRA	France	A	Annual	PREVM
156	I	FRA	France	A	Annual	PREVM
157	I	FRA	France	A	Annual	PREVM
158	I	DEU	Germany	A	Annual	PREVM
159	I	DEU	Germany	A	Annual	PREVM
160	I	DEU	Germany	A	Annual	PREVM
161	I	DEU	Germany	A	Annual	PREVM
162	I	DEU	Germany	A	Annual	PREVM
163	I	DEU	Germany	A	Annual	PREVM
164	I	DEU	Germany	A	Annual	PREVM
165	I	DEU	Germany	A	Annual	PREVM
166	I	DEU	Germany	A	Annual	PREVM
167	I	DEU	Germany	A	Annual	PREVM
168	I	DEU	Germany	A	Annual	PREVM
169	I	GRC	Greece	A	Annual	PREVM
170	I	GRC	Greece	A	Annual	PREVM
171	I	GRC	Greece	A	Annual	PREVM
172	I	GRC	Greece	A	Annual	PREVM
173	I	GRC	Greece	A	Annual	PREVM
174	I	GRC	Greece	A	Annual	PREVM
175	I	GRC	Greece	A	Annual	PREVM
176	I	GRC	Greece	A	Annual	PREVM
177	I	GRC	Greece	A	Annual	PREVM
178	I	HUN	Hungary	A	Annual	PREVM
179	I	HUN	Hungary	A	Annual	PREVM

180	I	HUN	Hungary	A	Annual	PREVM
181	I	HUN	Hungary	A	Annual	PREVM
182	I	HUN	Hungary	A	Annual	PREVM
183	I	HUN	Hungary	A	Annual	PREVM
184	I	HUN	Hungary	A	Annual	PREVM
185	I	HUN	Hungary	A	Annual	PREVM
186	I	HUN	Hungary	A	Annual	PREVM
187	I	HUN	Hungary	A	Annual	PREVM
188	I	HUN	Hungary	A	Annual	PREVM
189	I	HUN	Hungary	A	Annual	PREVM
190	I	HUN	Hungary	A	Annual	PREVM
191	I	HUN	Hungary	A	Annual	PREVM
192	I	ISL	Iceland	A	Annual	PREVM
193	I	ISL	Iceland	A	Annual	PREVM
194	I	ISL	Iceland	A	Annual	PREVM
195	I	ISL	Iceland	A	Annual	PREVM
196	I	ISL	Iceland	A	Annual	PREVM
197	I	ISL	Iceland	A	Annual	PREVM
198	I	ISL	Iceland	A	Annual	PREVM
199	I	ISL	Iceland	A	Annual	PREVM
200	I	ISL	Iceland	A	Annual	PREVM
201	I	ISL	Iceland	A	Annual	PREVM
202	I	ISL	Iceland	A	Annual	PREVM
203	I	ISL	Iceland	A	Annual	PREVM
204	I	ISL	Iceland	A	Annual	PREVM
205	I	IRL	Ireland	A	Annual	PREVM
206	I	IRL	Ireland	A	Annual	PREVM
207	I	IRL	Ireland	A	Annual	PREVM
208	I	IRL	Ireland	A	Annual	PREVM
209	I	IRL	Ireland	A	Annual	PREVM
210	I	IRL	Ireland	A	Annual	PREVM
211	I	IRL	Ireland	A	Annual	PREVM
212	I	IRL	Ireland	A	Annual	PREVM
213	I	IRL	Ireland	A	Annual	PREVM
214	I	IRL	Ireland	A	Annual	PREVM
215	I	IRL	Ireland	A	Annual	PREVM
216	I	IRL	Ireland	A	Annual	PREVM
217	I	IRL	Ireland	A	Annual	PREVM
218	I	ISR	Israel	A	Annual	PREVM
219	I	ISR	Israel	A	Annual	PREVM
220	I	ISR	Israel	A	Annual	PREVM
221	I	ISR	Israel	A	Annual	PREVM
222	I	ISR	Israel	A	Annual	PREVM
223	I	ISR	Israel	A	Annual	PREVM
224	I	ISR	Israel	A	Annual	PREVM
225	I	ISR	Israel	A	Annual	PREVM
226	I	ISR	Israel	A	Annual	PREVM
227	I	ISR	Israel	A	Annual	PREVM
228	I	ISR	Israel	A	Annual	PREVM
229	I	ISR	Israel	A	Annual	PREVM
230	I	ISR	Israel	A	Annual	PREVM

231	I	ITA	Italy	A	Annual	PREVM
232	I	ITA	Italy	A	Annual	PREVM
233	I	ITA	Italy	A	Annual	PREVM
234	I	ITA	Italy	A	Annual	PREVM
235	I	ITA	Italy	A	Annual	PREVM
236	I	ITA	Italy	A	Annual	PREVM
237	I	ITA	Italy	A	Annual	PREVM
238	I	ITA	Italy	A	Annual	PREVM
239	I	ITA	Italy	A	Annual	PREVM
240	I	ITA	Italy	A	Annual	PREVM
241	I	ITA	Italy	A	Annual	PREVM
242	I	ITA	Italy	A	Annual	PREVM
243	I	ITA	Italy	A	Annual	PREVM
244	I	JPN	Japan	A	Annual	PREVM
245	I	JPN	Japan	A	Annual	PREVM
246	I	JPN	Japan	A	Annual	PREVM
247	I	JPN	Japan	A	Annual	PREVM
248	I	JPN	Japan	A	Annual	PREVM
249	I	JPN	Japan	A	Annual	PREVM
250	I	JPN	Japan	A	Annual	PREVM
251	I	JPN	Japan	A	Annual	PREVM
252	I	JPN	Japan	A	Annual	PREVM
253	I	JPN	Japan	A	Annual	PREVM
254	I	JPN	Japan	A	Annual	PREVM
255	I	JPN	Japan	A	Annual	PREVM
256	I	KOR	Korea	A	Annual	PREVM
257	I	KOR	Korea	A	Annual	PREVM
258	I	KOR	Korea	A	Annual	PREVM
259	I	KOR	Korea	A	Annual	PREVM
260	I	KOR	Korea	A	Annual	PREVM
261	I	KOR	Korea	A	Annual	PREVM
262	I	KOR	Korea	A	Annual	PREVM
263	I	KOR	Korea	A	Annual	PREVM
264	I	KOR	Korea	A	Annual	PREVM
265	I	KOR	Korea	A	Annual	PREVM
266	I	KOR	Korea	A	Annual	PREVM
267	I	KOR	Korea	A	Annual	PREVM
268	I	KOR	Korea	A	Annual	PREVM
269	I	LVA	Latvia	A	Annual	PREVM
270	I	LVA	Latvia	A	Annual	PREVM
271	I	LVA	Latvia	A	Annual	PREVM
272	I	LVA	Latvia	A	Annual	PREVM
273	I	LVA	Latvia	A	Annual	PREVM
274	I	LVA	Latvia	A	Annual	PREVM
275	I	LVA	Latvia	A	Annual	PREVM
276	I	LVA	Latvia	A	Annual	PREVM
277	I	LVA	Latvia	A	Annual	PREVM
278	I	LVA	Latvia	A	Annual	PREVM
279	I	LVA	Latvia	A	Annual	PREVM
280	I	LVA	Latvia	A	Annual	PREVM
281	I	LVA	Latvia	A	Annual	PREVM

282	I	LVA	Latvia	A	Annual	PREVM
283	I	LTU	Lithuania	A	Annual	PREVM
284	I	LTU	Lithuania	A	Annual	PREVM
285	I	LTU	Lithuania	A	Annual	PREVM
286	I	LTU	Lithuania	A	Annual	PREVM
287	I	LTU	Lithuania	A	Annual	PREVM
288	I	LTU	Lithuania	A	Annual	PREVM
289	I	LTU	Lithuania	A	Annual	PREVM
290	I	LTU	Lithuania	A	Annual	PREVM
291	I	LTU	Lithuania	A	Annual	PREVM
292	I	LTU	Lithuania	A	Annual	PREVM
293	I	LTU	Lithuania	A	Annual	PREVM
294	I	LTU	Lithuania	A	Annual	PREVM
295	I	LTU	Lithuania	A	Annual	PREVM
296	I	LTU	Lithuania	A	Annual	PREVM
297	I	LUX	Luxembourg	A	Annual	PREVM
298	I	LUX	Luxembourg	A	Annual	PREVM
299	I	LUX	Luxembourg	A	Annual	PREVM
300	I	LUX	Luxembourg	A	Annual	PREVM
301	I	LUX	Luxembourg	A	Annual	PREVM
302	I	LUX	Luxembourg	A	Annual	PREVM
303	I	LUX	Luxembourg	A	Annual	PREVM
304	I	LUX	Luxembourg	A	Annual	PREVM
305	I	LUX	Luxembourg	A	Annual	PREVM
306	I	LUX	Luxembourg	A	Annual	PREVM
307	I	LUX	Luxembourg	A	Annual	PREVM
308	I	LUX	Luxembourg	A	Annual	PREVM
309	I	LUX	Luxembourg	A	Annual	PREVM
310	I	LUX	Luxembourg	A	Annual	PREVM
311	I	MEX	Mexico	A	Annual	PREVM
312	I	MEX	Mexico	A	Annual	PREVM
313	I	MEX	Mexico	A	Annual	PREVM
314	I	MEX	Mexico	A	Annual	PREVM
315	I	MEX	Mexico	A	Annual	PREVM
316	I	MEX	Mexico	A	Annual	PREVM
317	I	MEX	Mexico	A	Annual	PREVM
318	I	MEX	Mexico	A	Annual	PREVM
319	I	MEX	Mexico	A	Annual	PREVM
320	I	MEX	Mexico	A	Annual	PREVM
321	I	MEX	Mexico	A	Annual	PREVM
322	I	MEX	Mexico	A	Annual	PREVM
323	I	MEX	Mexico	A	Annual	PREVM
324	I	NLD	Netherlands	A	Annual	PREVM
325	I	NLD	Netherlands	A	Annual	PREVM
326	I	NLD	Netherlands	A	Annual	PREVM
327	I	NLD	Netherlands	A	Annual	PREVM
328	I	NLD	Netherlands	A	Annual	PREVM
329	I	NLD	Netherlands	A	Annual	PREVM
330	I	NLD	Netherlands	A	Annual	PREVM
331	I	NLD	Netherlands	A	Annual	PREVM
332	I	NLD	Netherlands	A	Annual	PREVM

333	I	NLD	Netherlands	A	Annual	PREVM
334	I	NLD	Netherlands	A	Annual	PREVM
335	I	NLD	Netherlands	A	Annual	PREVM
336	I	NLD	Netherlands	A	Annual	PREVM
337	I	NLD	Netherlands	A	Annual	PREVM
338	I	NZL	New Zealand	A	Annual	PREVM
339	I	NZL	New Zealand	A	Annual	PREVM
340	I	NZL	New Zealand	A	Annual	PREVM
341	I	NZL	New Zealand	A	Annual	PREVM
342	I	NZL	New Zealand	A	Annual	PREVM
343	I	NZL	New Zealand	A	Annual	PREVM
344	I	NZL	New Zealand	A	Annual	PREVM
345	I	NZL	New Zealand	A	Annual	PREVM
346	I	NZL	New Zealand	A	Annual	PREVM
347	I	NOR	Norway	A	Annual	PREVM
348	I	NOR	Norway	A	Annual	PREVM
349	I	NOR	Norway	A	Annual	PREVM
350	I	NOR	Norway	A	Annual	PREVM
351	I	NOR	Norway	A	Annual	PREVM
352	I	NOR	Norway	A	Annual	PREVM
353	I	NOR	Norway	A	Annual	PREVM
354	I	POL	Poland	A	Annual	PREVM
355	I	POL	Poland	A	Annual	PREVM
356	I	POL	Poland	A	Annual	PREVM
357	I	POL	Poland	A	Annual	PREVM
358	I	POL	Poland	A	Annual	PREVM
359	I	POL	Poland	A	Annual	PREVM
360	I	POL	Poland	A	Annual	PREVM
361	I	POL	Poland	A	Annual	PREVM
362	I	POL	Poland	A	Annual	PREVM
363	I	POL	Poland	A	Annual	PREVM
364	I	POL	Poland	A	Annual	PREVM
365	I	POL	Poland	A	Annual	PREVM
366	I	POL	Poland	A	Annual	PREVM
367	I	PRT	Portugal	A	Annual	PREVM
368	I	PRT	Portugal	A	Annual	PREVM
369	I	PRT	Portugal	A	Annual	PREVM
370	I	PRT	Portugal	A	Annual	PREVM
371	I	PRT	Portugal	A	Annual	PREVM
372	I	PRT	Portugal	A	Annual	PREVM
373	I	PRT	Portugal	A	Annual	PREVM
374	I	PRT	Portugal	A	Annual	PREVM
375	I	PRT	Portugal	A	Annual	PREVM
376	I	PRT	Portugal	A	Annual	PREVM
377	I	PRT	Portugal	A	Annual	PREVM
378	I	SVK	Slovak Republic	A	Annual	PREVM
379	I	SVK	Slovak Republic	A	Annual	PREVM
380	I	SVK	Slovak Republic	A	Annual	PREVM
381	I	SVK	Slovak Republic	A	Annual	PREVM
382	I	SVK	Slovak Republic	A	Annual	PREVM
383	I	SVK	Slovak Republic	A	Annual	PREVM

384	I	SVK	Slovak Republic	A	Annual	PREVM
385	I	SVK	Slovak Republic	A	Annual	PREVM
386	I	SVK	Slovak Republic	A	Annual	PREVM
387	I	SVK	Slovak Republic	A	Annual	PREVM
388	I	SVK	Slovak Republic	A	Annual	PREVM
389	I	SVK	Slovak Republic	A	Annual	PREVM
390	I	SVN	Slovenia	A	Annual	PREVM
391	I	SVN	Slovenia	A	Annual	PREVM
392	I	SVN	Slovenia	A	Annual	PREVM
393	I	SVN	Slovenia	A	Annual	PREVM
394	I	SVN	Slovenia	A	Annual	PREVM
395	I	SVN	Slovenia	A	Annual	PREVM
396	I	SVN	Slovenia	A	Annual	PREVM
397	I	SVN	Slovenia	A	Annual	PREVM
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401	I	SVN	Slovenia	A	Annual	PREVM
402	I	SVN	Slovenia	A	Annual	PREVM
403	I	SVN	Slovenia	A	Annual	PREVM
404	I	ESP	Spain	A	Annual	PREVM
405	I	ESP	Spain	A	Annual	PREVM
406	I	ESP	Spain	A	Annual	PREVM
407	I	ESP	Spain	A	Annual	PREVM
408	I	ESP	Spain	A	Annual	PREVM
409	I	ESP	Spain	A	Annual	PREVM
410	I	ESP	Spain	A	Annual	PREVM
411	I	ESP	Spain	A	Annual	PREVM
412	I	ESP	Spain	A	Annual	PREVM
413	I	ESP	Spain	A	Annual	PREVM
414	I	ESP	Spain	A	Annual	PREVM
415	I	ESP	Spain	A	Annual	PREVM
416	I	ESP	Spain	A	Annual	PREVM
417	I	ESP	Spain	A	Annual	PREVM
418	I	SWE	Sweden	A	Annual	PREVM
419	I	SWE	Sweden	A	Annual	PREVM
420	I	SWE	Sweden	A	Annual	PREVM
421	I	SWE	Sweden	A	Annual	PREVM
422	I	SWE	Sweden	A	Annual	PREVM
423	I	SWE	Sweden	A	Annual	PREVM
424	I	SWE	Sweden	A	Annual	PREVM
425	I	SWE	Sweden	A	Annual	PREVM
426	I	SWE	Sweden	A	Annual	PREVM
427	I	SWE	Sweden	A	Annual	PREVM
428	I	SWE	Sweden	A	Annual	PREVM
429	I	SWE	Sweden	A	Annual	PREVM
430	I	SWE	Sweden	A	Annual	PREVM
431	I	SWE	Sweden	A	Annual	PREVM
432	I	CHE	Switzerland	A	Annual	PREVM
433	I	CHE	Switzerland	A	Annual	PREVM
434	I	CHE	Switzerland	A	Annual	PREVM

435	I	CHE	Switzerland	A	Annual	PREVM
436	I	CHE	Switzerland	A	Annual	PREVM
437	I	CHE	Switzerland	A	Annual	PREVM
438	I	CHE	Switzerland	A	Annual	PREVM
439	I	CHE	Switzerland	A	Annual	PREVM
440	I	CHE	Switzerland	A	Annual	PREVM
441	I	CHE	Switzerland	A	Annual	PREVM
442	I	CHE	Switzerland	A	Annual	PREVM
443	I	CHE	Switzerland	A	Annual	PREVM
444	I	CHE	Switzerland	A	Annual	PREVM
445	I	CHE	Switzerland	A	Annual	PREVM
446	I	TUR	Türkiye	A	Annual	PREVM
447	I	TUR	Türkiye	A	Annual	PREVM
448	I	TUR	Türkiye	A	Annual	PREVM
449	I	TUR	Türkiye	A	Annual	PREVM
450	I	TUR	Türkiye	A	Annual	PREVM
451	I	TUR	Türkiye	A	Annual	PREVM
452	I	TUR	Türkiye	A	Annual	PREVM
453	I	TUR	Türkiye	A	Annual	PREVM
454	I	TUR	Türkiye	A	Annual	PREVM
455	I	TUR	Türkiye	A	Annual	PREVM
456	I	TUR	Türkiye	A	Annual	PREVM
457	I	TUR	Türkiye	A	Annual	PREVM
458	I	TUR	Türkiye	A	Annual	PREVM
459	I	TUR	Türkiye	A	Annual	PREVM
460	I	GBR	United Kingdom	A	Annual	PREVM
461	I	GBR	United Kingdom	A	Annual	PREVM
462	I	GBR	United Kingdom	A	Annual	PREVM
463	I	GBR	United Kingdom	A	Annual	PREVM
464	I	GBR	United Kingdom	A	Annual	PREVM
465	I	GBR	United Kingdom	A	Annual	PREVM
466	I	GBR	United Kingdom	A	Annual	PREVM
467	I	GBR	United Kingdom	A	Annual	PREVM
468	I	GBR	United Kingdom	A	Annual	PREVM
469	I	GBR	United Kingdom	A	Annual	PREVM
470	I	GBR	United Kingdom	A	Annual	PREVM
471	I	GBR	United Kingdom	A	Annual	PREVM
472	I	USA	United States	A	Annual	PREVM
473	I	USA	United States	A	Annual	PREVM
474	I	USA	United States	A	Annual	PREVM
475	I	USA	United States	A	Annual	PREVM
476	I	USA	United States	A	Annual	PREVM
477	I	USA	United States	A	Annual	PREVM
478	I	USA	United States	A	Annual	PREVM
479	I	USA	United States	A	Annual	PREVM
480	I	USA	United States	A	Annual	PREVM
481	I	USA	United States	A	Annual	PREVM
482	I	USA	United States	A	Annual	PREVM
483	I	USA	United States	A	Annual	PREVM
484	I	USA	United States	A	Annual	PREVM
485	I	ARG	Argentina	A	Annual	PREVM

486	I	ARG	Argentina	A	Annual	PREVM
487	I	ARG	Argentina	A	Annual	PREVM
488	I	ARG	Argentina	A	Annual	PREVM
489	I	ARG	Argentina	A	Annual	PREVM
490	I	ARG	Argentina	A	Annual	PREVM
491	I	ARG	Argentina	A	Annual	PREVM
492	I	ARG	Argentina	A	Annual	PREVM
493	I	ARG	Argentina	A	Annual	PREVM
494	I	ARG	Argentina	A	Annual	PREVM
495	I	ARG	Argentina	A	Annual	PREVM
496	I	ARG	Argentina	A	Annual	PREVM
497	I	ARG	Argentina	A	Annual	PREVM
498	I	BRA	Brazil	A	Annual	PREVM
499	I	BRA	Brazil	A	Annual	PREVM
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507	I	BRA	Brazil	A	Annual	PREVM
508	I	BRA	Brazil	A	Annual	PREVM
509	I	BRA	Brazil	A	Annual	PREVM
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521	I	BGR	Bulgaria	A	Annual	PREVM
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523	I	HRV	Croatia	A	Annual	PREVM
524	I	HRV	Croatia	A	Annual	PREVM
525	I	HRV	Croatia	A	Annual	PREVM
526	I	HRV	Croatia	A	Annual	PREVM
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533	I	HRV	Croatia	A	Annual	PREVM
534	I	HRV	Croatia	A	Annual	PREVM
535	I	PER	Peru	A	Annual	PREVM
536	I	PER	Peru	A	Annual	PREVM

537	I	PER	Peru	A	Annual	PREVM
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541	I	PER	Peru	A	Annual	PREVM
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543	I	PER	Peru	A	Annual	PREVM
544	I	PER	Peru	A	Annual	PREVM
545	I	PER	Peru	A	Annual	PREVM
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547	I	PER	Peru	A	Annual	PREVM
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560	I	ZAF	South Africa	A	Annual	PREVM
561	I	ZAF	South Africa	A	Annual	PREVM
562	I	ZAF	South Africa	A	Annual	PREVM
563	I	ZAF	South Africa	A	Annual	PREVM
564	I	ZAF	South Africa	A	Annual	PREVM
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568	I	ZAF	South Africa	A	Annual	PREVM
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570	I	ZAF	South Africa	A	Annual	PREVM
571	I	THA	Thailand	A	Annual	PREVM
572	I	THA	Thailand	A	Annual	PREVM
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580	I	THA	Thailand	A	Annual	PREVM
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582	I	AUS	Australia	A	Annual	TRTM
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587	I	AUS	Australia	A	Annual	TRTM

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620	I	BEL	Belgium	A	Annual	TRTM
621	I	BEL	Belgium	A	Annual	TRTM
622	I	CAN	Canada	A	Annual	TRTM
623	I	CAN	Canada	A	Annual	TRTM
624	I	CAN	Canada	A	Annual	TRTM
625	I	CAN	Canada	A	Annual	TRTM
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635	I	CHL	Chile	A	Annual	TRTM
636	I	CHL	Chile	A	Annual	TRTM
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687	I	DNK	Denmark	A	Annual	TRTM
688	I	DNK	Denmark	A	Annual	TRTM
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690	I	DNK	Denmark	A	Annual	TRTM
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712	I	EST	Estonia	A	Annual	TRTM
713	I	FIN	Finland	A	Annual	TRTM
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718	I	FIN	Finland	A	Annual	TRTM
719	I	FIN	Finland	A	Annual	TRTM
720	I	FIN	Finland	A	Annual	TRTM
721	I	FIN	Finland	A	Annual	TRTM
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725	I	FIN	Finland	A	Annual	TRTM
726	I	FRA	France	A	Annual	TRTM
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738	I	FRA	France	A	Annual	TRTM
739	I	DEU	Germany	A	Annual	TRTM
740	I	DEU	Germany	A	Annual	TRTM

741	I	DEU	Germany	A	Annual	TRTM
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743	I	DEU	Germany	A	Annual	TRTM
744	I	DEU	Germany	A	Annual	TRTM
745	I	DEU	Germany	A	Annual	TRTM
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750	I	GRC	Greece	A	Annual	TRTM
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752	I	GRC	Greece	A	Annual	TRTM
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754	I	GRC	Greece	A	Annual	TRTM
755	I	GRC	Greece	A	Annual	TRTM
756	I	GRC	Greece	A	Annual	TRTM
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790	I	IRL	Ireland	A	Annual	TRTM
791	I	IRL	Ireland	A	Annual	TRTM

792	I	IRL	Ireland	A	Annual	TRTM
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794	I	IRL	Ireland	A	Annual	TRTM
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796	I	IRL	Ireland	A	Annual	TRTM
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836	I	JPN	Japan	A	Annual	TRTM
837	I	KOR	Korea	A	Annual	TRTM
838	I	KOR	Korea	A	Annual	TRTM
839	I	KOR	Korea	A	Annual	TRTM
840	I	KOR	Korea	A	Annual	TRTM
841	I	KOR	Korea	A	Annual	TRTM
842	I	KOR	Korea	A	Annual	TRTM

843	I	KOR	Korea	A	Annual	TRTM
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845	I	KOR	Korea	A	Annual	TRTM
846	I	KOR	Korea	A	Annual	TRTM
847	I	KOR	Korea	A	Annual	TRTM
848	I	KOR	Korea	A	Annual	TRTM
849	I	KOR	Korea	A	Annual	TRTM
850	I	LVA	Latvia	A	Annual	TRTM
851	I	LVA	Latvia	A	Annual	TRTM
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871	I	LTU	Lithuania	A	Annual	TRTM
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873	I	LTU	Lithuania	A	Annual	TRTM
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877	I	LTU	Lithuania	A	Annual	TRTM
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885	I	LUX	Luxembourg	A	Annual	TRTM
886	I	LUX	Luxembourg	A	Annual	TRTM
887	I	LUX	Luxembourg	A	Annual	TRTM
888	I	LUX	Luxembourg	A	Annual	TRTM
889	I	LUX	Luxembourg	A	Annual	TRTM
890	I	LUX	Luxembourg	A	Annual	TRTM
891	I	LUX	Luxembourg	A	Annual	TRTM
892	I	MEX	Mexico	A	Annual	TRTM
893	I	MEX	Mexico	A	Annual	TRTM

894	I	MEX	Mexico	A	Annual	TRTM
895	I	MEX	Mexico	A	Annual	TRTM
896	I	MEX	Mexico	A	Annual	TRTM
897	I	MEX	Mexico	A	Annual	TRTM
898	I	MEX	Mexico	A	Annual	TRTM
899	I	MEX	Mexico	A	Annual	TRTM
900	I	MEX	Mexico	A	Annual	TRTM
901	I	MEX	Mexico	A	Annual	TRTM
902	I	MEX	Mexico	A	Annual	TRTM
903	I	MEX	Mexico	A	Annual	TRTM
904	I	MEX	Mexico	A	Annual	TRTM
905	I	NLD	Netherlands	A	Annual	TRTM
906	I	NLD	Netherlands	A	Annual	TRTM
907	I	NLD	Netherlands	A	Annual	TRTM
908	I	NLD	Netherlands	A	Annual	TRTM
909	I	NLD	Netherlands	A	Annual	TRTM
910	I	NLD	Netherlands	A	Annual	TRTM
911	I	NLD	Netherlands	A	Annual	TRTM
912	I	NLD	Netherlands	A	Annual	TRTM
913	I	NLD	Netherlands	A	Annual	TRTM
914	I	NLD	Netherlands	A	Annual	TRTM
915	I	NLD	Netherlands	A	Annual	TRTM
916	I	NLD	Netherlands	A	Annual	TRTM
917	I	NLD	Netherlands	A	Annual	TRTM
918	I	NLD	Netherlands	A	Annual	TRTM
919	I	NZL	New Zealand	A	Annual	TRTM
920	I	NZL	New Zealand	A	Annual	TRTM
921	I	NZL	New Zealand	A	Annual	TRTM
922	I	NZL	New Zealand	A	Annual	TRTM
923	I	NZL	New Zealand	A	Annual	TRTM
924	I	NZL	New Zealand	A	Annual	TRTM
925	I	NZL	New Zealand	A	Annual	TRTM
926	I	NZL	New Zealand	A	Annual	TRTM
927	I	NZL	New Zealand	A	Annual	TRTM
928	I	NOR	Norway	A	Annual	TRTM
929	I	NOR	Norway	A	Annual	TRTM
930	I	NOR	Norway	A	Annual	TRTM
931	I	NOR	Norway	A	Annual	TRTM
932	I	NOR	Norway	A	Annual	TRTM
933	I	NOR	Norway	A	Annual	TRTM
934	I	NOR	Norway	A	Annual	TRTM
935	I	POL	Poland	A	Annual	TRTM
936	I	POL	Poland	A	Annual	TRTM
937	I	POL	Poland	A	Annual	TRTM
938	I	POL	Poland	A	Annual	TRTM
939	I	POL	Poland	A	Annual	TRTM
940	I	POL	Poland	A	Annual	TRTM
941	I	POL	Poland	A	Annual	TRTM
942	I	POL	Poland	A	Annual	TRTM
943	I	POL	Poland	A	Annual	TRTM
944	I	POL	Poland	A	Annual	TRTM

945	I	POL	Poland	A	Annual	TRTM
946	I	POL	Poland	A	Annual	TRTM
947	I	POL	Poland	A	Annual	TRTM
948	I	PRT	Portugal	A	Annual	TRTM
949	I	PRT	Portugal	A	Annual	TRTM
950	I	PRT	Portugal	A	Annual	TRTM
951	I	PRT	Portugal	A	Annual	TRTM
952	I	PRT	Portugal	A	Annual	TRTM
953	I	PRT	Portugal	A	Annual	TRTM
954	I	PRT	Portugal	A	Annual	TRTM
955	I	PRT	Portugal	A	Annual	TRTM
956	I	PRT	Portugal	A	Annual	TRTM
957	I	PRT	Portugal	A	Annual	TRTM
958	I	PRT	Portugal	A	Annual	TRTM
959	I	SVK	Slovak Republic	A	Annual	TRTM
960	I	SVK	Slovak Republic	A	Annual	TRTM
961	I	SVK	Slovak Republic	A	Annual	TRTM
962	I	SVK	Slovak Republic	A	Annual	TRTM
963	I	SVK	Slovak Republic	A	Annual	TRTM
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968	I	SVK	Slovak Republic	A	Annual	TRTM
969	I	SVK	Slovak Republic	A	Annual	TRTM
970	I	SVK	Slovak Republic	A	Annual	TRTM
971	I	SVN	Slovenia	A	Annual	TRTM
972	I	SVN	Slovenia	A	Annual	TRTM
973	I	SVN	Slovenia	A	Annual	TRTM
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983	I	SVN	Slovenia	A	Annual	TRTM
984	I	SVN	Slovenia	A	Annual	TRTM
985	I	ESP	Spain	A	Annual	TRTM
986	I	ESP	Spain	A	Annual	TRTM
987	I	ESP	Spain	A	Annual	TRTM
988	I	ESP	Spain	A	Annual	TRTM
989	I	ESP	Spain	A	Annual	TRTM
990	I	ESP	Spain	A	Annual	TRTM
991	I	ESP	Spain	A	Annual	TRTM
992	I	ESP	Spain	A	Annual	TRTM
993	I	ESP	Spain	A	Annual	TRTM
994	I	ESP	Spain	A	Annual	TRTM
995	I	ESP	Spain	A	Annual	TRTM

996	I	ESP	Spain	A	Annual	TRTM
997	I	ESP	Spain	A	Annual	TRTM
998	I	ESP	Spain	A	Annual	TRTM
999	I	SWE	Sweden	A	Annual	TRTM
1000	I	SWE	Sweden	A	Annual	TRTM
1001	I	SWE	Sweden	A	Annual	TRTM
1002	I	SWE	Sweden	A	Annual	TRTM
1003	I	SWE	Sweden	A	Annual	TRTM
1004	I	SWE	Sweden	A	Annual	TRTM
1005	I	SWE	Sweden	A	Annual	TRTM
1006	I	SWE	Sweden	A	Annual	TRTM
1007	I	SWE	Sweden	A	Annual	TRTM
1008	I	SWE	Sweden	A	Annual	TRTM
1009	I	SWE	Sweden	A	Annual	TRTM
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1013	I	CHE	Switzerland	A	Annual	TRTM
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1016	I	CHE	Switzerland	A	Annual	TRTM
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1018	I	CHE	Switzerland	A	Annual	TRTM
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1020	I	CHE	Switzerland	A	Annual	TRTM
1021	I	CHE	Switzerland	A	Annual	TRTM
1022	I	CHE	Switzerland	A	Annual	TRTM
1023	I	CHE	Switzerland	A	Annual	TRTM
1024	I	CHE	Switzerland	A	Annual	TRTM
1025	I	CHE	Switzerland	A	Annual	TRTM
1026	I	CHE	Switzerland	A	Annual	TRTM
1027	I	TUR	Türkiye	A	Annual	TRTM
1028	I	TUR	Türkiye	A	Annual	TRTM
1029	I	TUR	Türkiye	A	Annual	TRTM
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1036	I	TUR	Türkiye	A	Annual	TRTM
1037	I	TUR	Türkiye	A	Annual	TRTM
1038	I	TUR	Türkiye	A	Annual	TRTM
1039	I	TUR	Türkiye	A	Annual	TRTM
1040	I	TUR	Türkiye	A	Annual	TRTM
1041	I	GBR	United Kingdom	A	Annual	TRTM
1042	I	GBR	United Kingdom	A	Annual	TRTM
1043	I	GBR	United Kingdom	A	Annual	TRTM
1044	I	GBR	United Kingdom	A	Annual	TRTM
1045	I	GBR	United Kingdom	A	Annual	TRTM
1046	I	GBR	United Kingdom	A	Annual	TRTM

1047	I	GBR	United Kingdom	A	Annual	TRTM
1048	I	GBR	United Kingdom	A	Annual	TRTM
1049	I	GBR	United Kingdom	A	Annual	TRTM
1050	I	GBR	United Kingdom	A	Annual	TRTM
1051	I	GBR	United Kingdom	A	Annual	TRTM
1052	I	GBR	United Kingdom	A	Annual	TRTM
1053	I	USA	United States	A	Annual	TRTM
1054	I	USA	United States	A	Annual	TRTM
1055	I	USA	United States	A	Annual	TRTM
1056	I	USA	United States	A	Annual	TRTM
1057	I	USA	United States	A	Annual	TRTM
1058	I	USA	United States	A	Annual	TRTM
1059	I	USA	United States	A	Annual	TRTM
1060	I	USA	United States	A	Annual	TRTM
1061	I	USA	United States	A	Annual	TRTM
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1063	I	USA	United States	A	Annual	TRTM
1064	I	USA	United States	A	Annual	TRTM
1065	I	USA	United States	A	Annual	TRTM
1066	I	ARG	Argentina	A	Annual	TRTM
1067	I	ARG	Argentina	A	Annual	TRTM
1068	I	ARG	Argentina	A	Annual	TRTM
1069	I	ARG	Argentina	A	Annual	TRTM
1070	I	ARG	Argentina	A	Annual	TRTM
1071	I	ARG	Argentina	A	Annual	TRTM
1072	I	ARG	Argentina	A	Annual	TRTM
1073	I	ARG	Argentina	A	Annual	TRTM
1074	I	ARG	Argentina	A	Annual	TRTM
1075	I	ARG	Argentina	A	Annual	TRTM
1076	I	ARG	Argentina	A	Annual	TRTM
1077	I	ARG	Argentina	A	Annual	TRTM
1078	I	ARG	Argentina	A	Annual	TRTM
1079	I	BRA	Brazil	A	Annual	TRTM
1080	I	BRA	Brazil	A	Annual	TRTM
1081	I	BRA	Brazil	A	Annual	TRTM
1082	I	BRA	Brazil	A	Annual	TRTM
1083	I	BRA	Brazil	A	Annual	TRTM
1084	I	BRA	Brazil	A	Annual	TRTM
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1086	I	BRA	Brazil	A	Annual	TRTM
1087	I	BRA	Brazil	A	Annual	TRTM
1088	I	BRA	Brazil	A	Annual	TRTM
1089	I	BRA	Brazil	A	Annual	TRTM
1090	I	BRA	Brazil	A	Annual	TRTM
1091	I	BGR	Bulgaria	A	Annual	TRTM
1092	I	BGR	Bulgaria	A	Annual	TRTM
1093	I	BGR	Bulgaria	A	Annual	TRTM
1094	I	BGR	Bulgaria	A	Annual	TRTM
1095	I	BGR	Bulgaria	A	Annual	TRTM
1096	I	BGR	Bulgaria	A	Annual	TRTM
1097	I	BGR	Bulgaria	A	Annual	TRTM

1098	I	BGR	Bulgaria	A	Annual	TRTM
1099	I	BGR	Bulgaria	A	Annual	TRTM
1100	I	BGR	Bulgaria	A	Annual	TRTM
1101	I	BGR	Bulgaria	A	Annual	TRTM
1102	I	BGR	Bulgaria	A	Annual	TRTM
1103	I	BGR	Bulgaria	A	Annual	TRTM
1104	I	HRV	Croatia	A	Annual	TRTM
1105	I	HRV	Croatia	A	Annual	TRTM
1106	I	HRV	Croatia	A	Annual	TRTM
1107	I	HRV	Croatia	A	Annual	TRTM
1108	I	HRV	Croatia	A	Annual	TRTM
1109	I	HRV	Croatia	A	Annual	TRTM
1110	I	HRV	Croatia	A	Annual	TRTM
1111	I	HRV	Croatia	A	Annual	TRTM
1112	I	HRV	Croatia	A	Annual	TRTM
1113	I	HRV	Croatia	A	Annual	TRTM
1114	I	HRV	Croatia	A	Annual	TRTM
1115	I	HRV	Croatia	A	Annual	TRTM
1116	I	PER	Peru	A	Annual	TRTM
1117	I	PER	Peru	A	Annual	TRTM
1118	I	PER	Peru	A	Annual	TRTM
1119	I	PER	Peru	A	Annual	TRTM
1120	I	PER	Peru	A	Annual	TRTM
1121	I	PER	Peru	A	Annual	TRTM
1122	I	PER	Peru	A	Annual	TRTM
1123	I	PER	Peru	A	Annual	TRTM
1124	I	PER	Peru	A	Annual	TRTM
1125	I	PER	Peru	A	Annual	TRTM
1126	I	PER	Peru	A	Annual	TRTM
1127	I	PER	Peru	A	Annual	TRTM
1128	I	PER	Peru	A	Annual	TRTM
1129	I	ROU	Romania	A	Annual	TRTM
1130	I	ROU	Romania	A	Annual	TRTM
1131	I	ROU	Romania	A	Annual	TRTM
1132	I	ROU	Romania	A	Annual	TRTM
1133	I	ROU	Romania	A	Annual	TRTM
1134	I	ROU	Romania	A	Annual	TRTM
1135	I	ROU	Romania	A	Annual	TRTM
1136	I	ROU	Romania	A	Annual	TRTM
1137	I	ROU	Romania	A	Annual	TRTM
1138	I	ROU	Romania	A	Annual	TRTM
1139	I	ROU	Romania	A	Annual	TRTM
1140	I	ROU	Romania	A	Annual	TRTM
1141	I	ZAF	South Africa	A	Annual	TRTM
1142	I	ZAF	South Africa	A	Annual	TRTM
1143	I	ZAF	South Africa	A	Annual	TRTM
1144	I	ZAF	South Africa	A	Annual	TRTM
1145	I	ZAF	South Africa	A	Annual	TRTM
1146	I	ZAF	South Africa	A	Annual	TRTM
1147	I	ZAF	South Africa	A	Annual	TRTM
1148	I	ZAF	South Africa	A	Annual	TRTM

1149	I	ZAF	South Africa	A	Annual	TRTM
1150	I	ZAF	South Africa	A	Annual	TRTM
1151	I	ZAF	South Africa	A	Annual	TRTM
1152	I	THA	Thailand	A	Annual	TRTM
1153	I	THA	Thailand	A	Annual	TRTM
1154	I	THA	Thailand	A	Annual	TRTM
1155	I	THA	Thailand	A	Annual	TRTM
1156	I	THA	Thailand	A	Annual	TRTM
1157	I	THA	Thailand	A	Annual	TRTM
1158	I	THA	Thailand	A	Annual	TRTM
1159	I	THA	Thailand	A	Annual	TRTM
1160	I	THA	Thailand	A	Annual	TRTM
1161	I	THA	Thailand	A	Annual	TRTM
1162	I	THA	Thailand	A	Annual	TRTM

		Measure	UNIT_MEASURE			Unit.of.measure	AGE
1	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
2	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
3	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
4	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
5	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
6	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
7	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
8	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
9	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
10	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
11	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
12	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
13	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
14	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
15	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
16	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
17	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
18	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
19	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
20	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
21	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
22	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
23	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
24	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
25	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
26	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
27	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
28	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
29	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
30	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
31	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
32	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
33	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
34	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
35	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T
36	Preventable mortality	DT_10P5HB	Deaths	per	100 000	inhabitants	_T

[illegible]

1159	Treatable mortality	DT_10P5HB	Deaths per 100 000 inhabitants	_T
1160	Treatable mortality	DT_10P5HB	Deaths per 100 000 inhabitants	_T
1161	Treatable mortality	DT_10P5HB	Deaths per 100 000 inhabitants	_T
1162	Treatable mortality	DT_10P5HB	Deaths per 100 000 inhabitants	_T
	Age SEX	Sex	SOCIO_ECON_STATUS	Socio.economic.status DEATH_CAUSE
1	Total	_T	Total	_Z Not applicable _Z
2	Total	_T	Total	_Z Not applicable _Z
3	Total	_T	Total	_Z Not applicable _Z
4	Total	_T	Total	_Z Not applicable _Z
5	Total	_T	Total	_Z Not applicable _Z
6	Total	_T	Total	_Z Not applicable _Z
7	Total	_T	Total	_Z Not applicable _Z
8	Total	_T	Total	_Z Not applicable _Z
9	Total	_T	Total	_Z Not applicable _Z
10	Total	_T	Total	_Z Not applicable _Z
11	Total	_T	Total	_Z Not applicable _Z
12	Total	_T	Total	_Z Not applicable _Z
13	Total	_T	Total	_Z Not applicable _Z
14	Total	_T	Total	_Z Not applicable _Z
15	Total	_T	Total	_Z Not applicable _Z
16	Total	_T	Total	_Z Not applicable _Z
17	Total	_T	Total	_Z Not applicable _Z
18	Total	_T	Total	_Z Not applicable _Z
19	Total	_T	Total	_Z Not applicable _Z
20	Total	_T	Total	_Z Not applicable _Z
21	Total	_T	Total	_Z Not applicable _Z
22	Total	_T	Total	_Z Not applicable _Z
23	Total	_T	Total	_Z Not applicable _Z
24	Total	_T	Total	_Z Not applicable _Z
25	Total	_T	Total	_Z Not applicable _Z
26	Total	_T	Total	_Z Not applicable _Z
27	Total	_T	Total	_Z Not applicable _Z
28	Total	_T	Total	_Z Not applicable _Z
29	Total	_T	Total	_Z Not applicable _Z
30	Total	_T	Total	_Z Not applicable _Z
31	Total	_T	Total	_Z Not applicable _Z
32	Total	_T	Total	_Z Not applicable _Z
33	Total	_T	Total	_Z Not applicable _Z
34	Total	_T	Total	_Z Not applicable _Z
35	Total	_T	Total	_Z Not applicable _Z
36	Total	_T	Total	_Z Not applicable _Z
37	Total	_T	Total	_Z Not applicable _Z
38	Total	_T	Total	_Z Not applicable _Z
39	Total	_T	Total	_Z Not applicable _Z
40	Total	_T	Total	_Z Not applicable _Z
41	Total	_T	Total	_Z Not applicable _Z
42	Total	_T	Total	_Z Not applicable _Z
43	Total	_T	Total	_Z Not applicable _Z
44	Total	_T	Total	_Z Not applicable _Z
45	Total	_T	Total	_Z Not applicable _Z
46	Total	_T	Total	_Z Not applicable _Z

47	Total	_T	Total	_Z	Not applicable	_Z
48	Total	_T	Total	_Z	Not applicable	_Z
49	Total	_T	Total	_Z	Not applicable	_Z
50	Total	_T	Total	_Z	Not applicable	_Z
51	Total	_T	Total	_Z	Not applicable	_Z
52	Total	_T	Total	_Z	Not applicable	_Z
53	Total	_T	Total	_Z	Not applicable	_Z
54	Total	_T	Total	_Z	Not applicable	_Z
55	Total	_T	Total	_Z	Not applicable	_Z
56	Total	_T	Total	_Z	Not applicable	_Z
57	Total	_T	Total	_Z	Not applicable	_Z
58	Total	_T	Total	_Z	Not applicable	_Z
59	Total	_T	Total	_Z	Not applicable	_Z
60	Total	_T	Total	_Z	Not applicable	_Z
61	Total	_T	Total	_Z	Not applicable	_Z
62	Total	_T	Total	_Z	Not applicable	_Z
63	Total	_T	Total	_Z	Not applicable	_Z
64	Total	_T	Total	_Z	Not applicable	_Z
65	Total	_T	Total	_Z	Not applicable	_Z
66	Total	_T	Total	_Z	Not applicable	_Z
67	Total	_T	Total	_Z	Not applicable	_Z
68	Total	_T	Total	_Z	Not applicable	_Z
69	Total	_T	Total	_Z	Not applicable	_Z
70	Total	_T	Total	_Z	Not applicable	_Z
71	Total	_T	Total	_Z	Not applicable	_Z
72	Total	_T	Total	_Z	Not applicable	_Z
73	Total	_T	Total	_Z	Not applicable	_Z
74	Total	_T	Total	_Z	Not applicable	_Z
75	Total	_T	Total	_Z	Not applicable	_Z
76	Total	_T	Total	_Z	Not applicable	_Z
77	Total	_T	Total	_Z	Not applicable	_Z
78	Total	_T	Total	_Z	Not applicable	_Z
79	Total	_T	Total	_Z	Not applicable	_Z
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12	Not applicable	_Z	Not applicable	_Z	Not applicable
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21	Not applicable	_Z	Not applicable	_Z	Not applicable
22	Not applicable	_Z	Not applicable	_Z	Not applicable
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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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[illegible]

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[illegible]

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2	2011	NA	109
3	2012	NA	105
4	2013	NA	105
5	2014	NA	107
6	2015	NA	108
7	2016	NA	103
8	2017	NA	103
9	2018	NA	101
10	2019	NA	104
11	2020	NA	97
12	2021	NA	97
13	2022	NA	107
14	2023	NA	99
15	2010	NA	144
16	2011	NA	140
17	2012	NA	140
18	2013	NA	136
19	2014	NA	134
20	2015	NA	137
21	2016	NA	132
22	2017	NA	128
23	2018	NA	128
24	2019	NA	124
25	2020	NA	134
26	2021	NA	141
27	2022	NA	132
28	2023	NA	121
29	2010	NA	147
30	2011	NA	144
31	2012	NA	142
32	2013	NA	138
33	2014	NA	133
34	2015	NA	133
35	2016	NA	127

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37	2018	NA	121	NA	1	One
38	2019	NA	117	NA	1	One
39	2020	NA	144	NA	1	One
40	2021	NA	134	NA	1	One
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45	2014	NA	127	NA	1	One
46	2015	NA	125	NA	1	One
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48	2017	NA	124	NA	1	One
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51	2020	NA	129	NA	1	One
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57	2013	NA	146	NA	1	One
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59	2015	NA	142	NA	1	One
60	2016	NA	136	NA	1	One
61	2017	NA	129	NA	1	One
62	2018	NA	121	NA	1	One
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65	2021	NA	190	NA	1	One
66	2022	NA	151	NA	1	One
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70	2013	NA	165	NA	1	One
71	2014	NA	159	NA	1	One
72	2015	NA	157	NA	1	One
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77	2020	NA	223	NA	1	One
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84	2015	NA	126	NA	1	One
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86	2017	NA	122	NA	1	One

87	2018	NA	126	NA	1	One
88	2019	NA	136	NA	1	One
89	2020	NA	149	NA	1	One
90	2021	NA	209	NA	1	One
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95	2013	NA	181	NA	1	One
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97	2015	NA	170	NA	1	One
98	2016	NA	160	NA	1	One
99	2017	NA	162	NA	1	One
100	2018	NA	160	NA	1	One
101	2019	NA	155	NA	1	One
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122	2013	NA	230	NA	1	One
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127	2018	NA	211	NA	1	One
128	2019	NA	192	NA	1	One
129	2020	NA	210	NA	1	One
130	2021	NA	253	NA	1	One
131	2022	NA	222	NA	1	One
132	2010	NA	160	NA	1	One
133	2011	NA	154	NA	1	One
134	2012	NA	150	NA	1	One
135	2013	NA	146	NA	1	One
136	2014	NA	138	NA	1	One
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184	2016	NA	271	NA	1	One
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186	2018	NA	271	NA	1	One
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188	2020	NA	290	NA	1	One

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190	2022	NA	276	NA	1	One
191	2023	NA	249	NA	1	One
192	2010	NA	116	NA	1	One
193	2011	NA	110	NA	1	One
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206	2011	NA	132	NA	1	One
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242	2021	NA	106	NA	1	One
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244	2010	NA	115	NA	1	One
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273	2014	NA	293	NA	1	One
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275	2016	NA	277	NA	1	One
276	2017	NA	273	NA	1	One
277	2018	NA	272	NA	1	One
278	2019	NA	248	NA	1	One
279	2020	NA	262	NA	1	One
280	2021	NA	364	NA	1	One
281	2022	NA	285	NA	1	One
282	2023	NA	258	NA	1	One
283	2010	NA	340	NA	1	One
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285	2012	NA	318	NA	1	One
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288	2015	NA	293	NA	1	One
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290	2017	NA	260	NA	1	One

291	2018	NA	247	NA	1	One
292	2019	NA	241	NA	1	One
293	2020	NA	285	NA	1	One
294	2021	NA	326	NA	1	One
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356	2012	NA	202	NA	1	One
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358	2014	NA	186	NA	1	One
359	2015	NA	184	NA	1	One
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361	2017	NA	182	NA	1	One
362	2018	NA	181	NA	1	One
363	2019	NA	182	NA	1	One
364	2020	NA	230	NA	1	One
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391	2011	NA	168	NA	1	One
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393	2013	NA	161	NA	1	One
394	2014	NA	151	NA	1	One
395	2015	NA	154	NA	1	One
396	2016	NA	151	NA	1	One
397	2017	NA	150	NA	1	One
398	2018	NA	145	NA	1	One
399	2019	NA	142	NA	1	One
400	2020	NA	164	NA	1	One
401	2021	NA	177	NA	1	One
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429	2021	NA	96	NA	1	One
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431	2023	NA	86	NA	1	One
432	2010	NA	104	NA	1	One
433	2011	NA	101	NA	1	One
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435	2013	NA	100	NA	1	One
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446	2010	NA	147	NA	1	One
447	2011	NA	148	NA	1	One
448	2012	NA	142	NA	1	One
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450	2014	NA	160	NA	1	One
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453	2017	NA	150	NA	1	One
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457	2021	NA	183	NA	1	One
458	2022	NA	127	NA	1	One
459	2023	NA	168	NA	1	One
460	2010	NA	137	NA	1	One
461	2011	NA	134	NA	1	One
462	2012	NA	129	NA	1	One
463	2013	NA	129	NA	1	One
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466	2016	NA	128	NA	1	One
467	2017	NA	125	NA	1	One
468	2018	NA	129	NA	1	One
469	2019	NA	123	NA	1	One
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471	2021	NA	156	NA	1	One
472	2010	NA	181	NA	1	One
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476	2014	NA	178	NA	1	One
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481	2019	NA	177	NA	1	One
482	2020	NA	232	NA	1	One
483	2021	NA	266	NA	1	One
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496	2021	NA	269	NA	1	One
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498	2010	NA	224	NA	1	One
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502	2014	NA	211	NA	1	One
503	2015	NA	206	NA	1	One
504	2016	NA	207	NA	1	One
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506	2018	NA	195	NA	1	One
507	2019	NA	189	NA	1	One
508	2020	NA	259	NA	1	One
509	2021	NA	356	NA	1	One
510	2010	NA	212	NA	1	One
511	2011	NA	201	NA	1	One
512	2012	NA	211	NA	1	One
513	2013	NA	198	NA	1	One
514	2014	NA	203	NA	1	One
515	2015	NA	203	NA	1	One
516	2016	NA	192	NA	1	One
517	2017	NA	190	NA	1	One
518	2018	NA	188	NA	1	One
519	2019	NA	192	NA	1	One
520	2020	NA	261	NA	1	One
521	2021	NA	378	NA	1	One
522	2022	NA	228	NA	1	One
523	2010	NA	221	NA	1	One
524	2011	NA	212	NA	1	One
525	2012	NA	210	NA	1	One
526	2013	NA	200	NA	1	One
527	2014	NA	196	NA	1	One
528	2015	NA	201	NA	1	One
529	2016	NA	191	NA	1	One
530	2017	NA	191	NA	1	One
531	2018	NA	196	NA	1	One
532	2019	NA	192	NA	1	One
533	2020	NA	216	NA	1	One
534	2021	NA	254	NA	1	One
535	2010	NA	106	NA	1	One
536	2011	NA	101	NA	1	One
537	2012	NA	102	NA	1	One
538	2013	NA	103	NA	1	One
539	2014	NA	97	NA	1	One
540	2015	NA	97	NA	1	One
541	2016	NA	86	NA	1	One
542	2017	NA	91	NA	1	One
543	2018	NA	107	NA	1	One
544	2019	NA	111	NA	1	One
545	2020	NA	408	NA	1	One

546	2021	NA	447	NA	1	One
547	2022	NA	146	NA	1	One
548	2010	NA	303	NA	1	One
549	2011	NA	276	NA	1	One
550	2012	NA	276	NA	1	One
551	2013	NA	263	NA	1	One
552	2014	NA	264	NA	1	One
553	2015	NA	260	NA	1	One
554	2016	NA	257	NA	1	One
555	2017	NA	254	NA	1	One
556	2018	NA	254	NA	1	One
557	2019	NA	245	NA	1	One
558	2020	NA	296	NA	1	One
559	2022	NA	251	NA	1	One
560	2010	NA	438	NA	1	One
561	2011	NA	399	NA	1	One
562	2012	NA	378	NA	1	One
563	2013	NA	365	NA	1	One
564	2014	NA	364	NA	1	One
565	2015	NA	360	NA	1	One
566	2016	NA	354	NA	1	One
567	2017	NA	344	NA	1	One
568	2018	NA	332	NA	1	One
569	2019	NA	318	NA	1	One
570	2020	NA	358	NA	1	One
571	2010	NA	166	NA	1	One
572	2011	NA	169	NA	1	One
573	2012	NA	170	NA	1	One
574	2013	NA	178	NA	1	One
575	2014	NA	178	NA	1	One
576	2015	NA	182	NA	1	One
577	2016	NA	189	NA	1	One
578	2017	NA	181	NA	1	One
579	2018	NA	176	NA	1	One
580	2019	NA	181	NA	1	One
581	2021	NA	199	NA	1	One
582	2010	NA	60	NA	1	One
583	2011	NA	59	NA	1	One
584	2012	NA	56	NA	1	One
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586	2014	NA	55	NA	1	One
587	2015	NA	54	NA	1	One
588	2016	NA	53	NA	1	One
589	2017	NA	51	NA	1	One
590	2018	NA	49	NA	1	One
591	2019	NA	51	NA	1	One
592	2020	NA	48	NA	1	One
593	2021	NA	47	NA	1	One
594	2022	NA	49	NA	1	One
595	2023	NA	47	NA	1	One
596	2010	NA	70	NA	1	One

597	2011	NA	66	NA	1	One
598	2012	NA	64	NA	1	One
599	2013	NA	63	NA	1	One
600	2014	NA	63	NA	1	One
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602	2016	NA	63	NA	1	One
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606	2020	NA	57	NA	1	One
607	2021	NA	57	NA	1	One
608	2022	NA	55	NA	1	One
609	2023	NA	54	NA	1	One
610	2010	NA	70	NA	1	One
611	2011	NA	68	NA	1	One
612	2012	NA	67	NA	1	One
613	2013	NA	66	NA	1	One
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615	2015	NA	60	NA	1	One
616	2016	NA	58	NA	1	One
617	2017	NA	57	NA	1	One
618	2018	NA	57	NA	1	One
619	2019	NA	53	NA	1	One
620	2020	NA	51	NA	1	One
621	2021	NA	50	NA	1	One
622	2010	NA	69	NA	1	One
623	2011	NA	67	NA	1	One
624	2012	NA	66	NA	1	One
625	2013	NA	65	NA	1	One
626	2014	NA	64	NA	1	One
627	2015	NA	63	NA	1	One
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629	2017	NA	60	NA	1	One
630	2018	NA	59	NA	1	One
631	2019	NA	59	NA	1	One
632	2020	NA	60	NA	1	One
633	2021	NA	57	NA	1	One
634	2022	NA	58	NA	1	One
635	2010	NA	95	NA	1	One
636	2011	NA	88	NA	1	One
637	2012	NA	88	NA	1	One
638	2013	NA	90	NA	1	One
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640	2015	NA	87	NA	1	One
641	2016	NA	83	NA	1	One
642	2017	NA	81	NA	1	One
643	2018	NA	79	NA	1	One
644	2019	NA	81	NA	1	One
645	2020	NA	75	NA	1	One
646	2021	NA	76	NA	1	One
647	2022	NA	78	NA	1	One

648	2010	NA	116	NA	1	One
649	2011	NA	109	NA	1	One
650	2012	NA	109	NA	1	One
651	2013	NA	107	NA	1	One
652	2014	NA	103	NA	1	One
653	2015	NA	105	NA	1	One
654	2016	NA	105	NA	1	One
655	2017	NA	101	NA	1	One
656	2018	NA	103	NA	1	One
657	2019	NA	99	NA	1	One
658	2020	NA	105	NA	1	One
659	2021	NA	115	NA	1	One
660	2010	NA	98	NA	1	One
661	2011	NA	91	NA	1	One
662	2012	NA	89	NA	1	One
663	2013	NA	87	NA	1	One
664	2014	NA	89	NA	1	One
665	2015	NA	87	NA	1	One
666	2016	NA	89	NA	1	One
667	2017	NA	90	NA	1	One
668	2018	NA	93	NA	1	One
669	2019	NA	103	NA	1	One
670	2020	NA	90	NA	1	One
671	2021	NA	93	NA	1	One
672	2022	NA	94	NA	1	One
673	2010	NA	130	NA	1	One
674	2011	NA	120	NA	1	One
675	2012	NA	118	NA	1	One
676	2013	NA	116	NA	1	One
677	2014	NA	107	NA	1	One
678	2015	NA	109	NA	1	One
679	2016	NA	104	NA	1	One
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681	2018	NA	100	NA	1	One
682	2019	NA	97	NA	1	One
683	2020	NA	99	NA	1	One
684	2021	NA	102	NA	1	One
685	2022	NA	91	NA	1	One
686	2023	NA	87	NA	1	One
687	2010	NA	78	NA	1	One
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689	2012	NA	69	NA	1	One
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693	2016	NA	64	NA	1	One
694	2017	NA	60	NA	1	One
695	2018	NA	59	NA	1	One
696	2019	NA	56	NA	1	One
697	2020	NA	54	NA	1	One
698	2021	NA	54	NA	1	One

699	2022	NA	55	NA	1	One
700	2010	NA	142	NA	1	One
701	2011	NA	131	NA	1	One
702	2012	NA	132	NA	1	One
703	2013	NA	122	NA	1	One
704	2014	NA	125	NA	1	One
705	2015	NA	116	NA	1	One
706	2016	NA	112	NA	1	One
707	2017	NA	111	NA	1	One
708	2018	NA	109	NA	1	One
709	2019	NA	104	NA	1	One
710	2020	NA	100	NA	1	One
711	2021	NA	110	NA	1	One
712	2022	NA	101	NA	1	One
713	2010	NA	74	NA	1	One
714	2011	NA	70	NA	1	One
715	2012	NA	68	NA	1	One
716	2013	NA	65	NA	1	One
717	2014	NA	62	NA	1	One
718	2015	NA	60	NA	1	One
719	2016	NA	62	NA	1	One
720	2017	NA	61	NA	1	One
721	2018	NA	58	NA	1	One
722	2019	NA	56	NA	1	One
723	2020	NA	57	NA	1	One
724	2021	NA	56	NA	1	One
725	2022	NA	57	NA	1	One
726	2010	NA	59	NA	1	One
727	2011	NA	56	NA	1	One
728	2012	NA	54	NA	1	One
729	2013	NA	53	NA	1	One
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731	2015	NA	51	NA	1	One
732	2016	NA	50	NA	1	One
733	2017	NA	50	NA	1	One
734	2018	NA	50	NA	1	One
735	2019	NA	49	NA	1	One
736	2020	NA	47	NA	1	One
737	2021	NA	47	NA	1	One
738	2022	NA	48	NA	1	One
739	2010	NA	78	NA	1	One
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742	2013	NA	74	NA	1	One
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744	2015	NA	73	NA	1	One
745	2016	NA	71	NA	1	One
746	2017	NA	69	NA	1	One
747	2018	NA	69	NA	1	One
748	2019	NA	66	NA	1	One
749	2020	NA	66	NA	1	One

750	2014	NA	79	NA	1	One
751	2015	NA	80	NA	1	One
752	2016	NA	80	NA	1	One
753	2017	NA	78	NA	1	One
754	2018	NA	75	NA	1	One
755	2019	NA	78	NA	1	One
756	2020	NA	76	NA	1	One
757	2021	NA	78	NA	1	One
758	2022	NA	72	NA	1	One
759	2010	NA	167	NA	1	One
760	2011	NA	163	NA	1	One
761	2012	NA	159	NA	1	One
762	2013	NA	152	NA	1	One
763	2014	NA	150	NA	1	One
764	2015	NA	151	NA	1	One
765	2016	NA	144	NA	1	One
766	2017	NA	147	NA	1	One
767	2018	NA	144	NA	1	One
768	2019	NA	142	NA	1	One
769	2020	NA	147	NA	1	One
770	2021	NA	155	NA	1	One
771	2022	NA	146	NA	1	One
772	2023	NA	141	NA	1	One
773	2010	NA	51	NA	1	One
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777	2014	NA	48	NA	1	One
778	2015	NA	51	NA	1	One
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781	2018	NA	51	NA	1	One
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783	2020	NA	48	NA	1	One
784	2021	NA	49	NA	1	One
785	2022	NA	49	NA	1	One
786	2010	NA	78	NA	1	One
787	2011	NA	77	NA	1	One
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796	2020	NA	56	NA	1	One
797	2021	NA	55	NA	1	One
798	2022	NA	57	NA	1	One
799	2010	NA	76	NA	1	One
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801	2012	NA	74	NA	1	One
802	2013	NA	71	NA	1	One
803	2014	NA	68	NA	1	One
804	2015	NA	70	NA	1	One
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806	2017	NA	63	NA	1	One
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810	2021	NA	57	NA	1	One
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812	2010	NA	63	NA	1	One
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814	2012	NA	62	NA	1	One
815	2013	NA	59	NA	1	One
816	2014	NA	57	NA	1	One
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818	2016	NA	55	NA	1	One
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820	2018	NA	54	NA	1	One
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822	2020	NA	55	NA	1	One
823	2021	NA	53	NA	1	One
824	2022	NA	52	NA	1	One
825	2010	NA	61	NA	1	One
826	2011	NA	61	NA	1	One
827	2012	NA	59	NA	1	One
828	2013	NA	58	NA	1	One
829	2014	NA	57	NA	1	One
830	2015	NA	55	NA	1	One
831	2016	NA	55	NA	1	One
832	2017	NA	52	NA	1	One
833	2018	NA	51	NA	1	One
834	2019	NA	51	NA	1	One
835	2020	NA	49	NA	1	One
836	2021	NA	49	NA	1	One
837	2010	NA	63	NA	1	One
838	2011	NA	60	NA	1	One
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840	2013	NA	55	NA	1	One
841	2014	NA	53	NA	1	One
842	2015	NA	52	NA	1	One
843	2016	NA	51	NA	1	One
844	2017	NA	48	NA	1	One
845	2018	NA	47	NA	1	One
846	2019	NA	44	NA	1	One
847	2020	NA	43	NA	1	One
848	2021	NA	44	NA	1	One
849	2022	NA	45	NA	1	One
850	2010	NA	203	NA	1	One
851	2011	NA	194	NA	1	One

852	2012	NA	185	NA	1	One
853	2013	NA	181	NA	1	One
854	2014	NA	171	NA	1	One
855	2015	NA	169	NA	1	One
856	2016	NA	166	NA	1	One
857	2017	NA	163	NA	1	One
858	2018	NA	160	NA	1	One
859	2019	NA	154	NA	1	One
860	2020	NA	151	NA	1	One
861	2021	NA	167	NA	1	One
862	2022	NA	164	NA	1	One
863	2023	NA	154	NA	1	One
864	2010	NA	187	NA	1	One
865	2011	NA	183	NA	1	One
866	2012	NA	181	NA	1	One
867	2013	NA	177	NA	1	One
868	2014	NA	167	NA	1	One
869	2015	NA	176	NA	1	One
870	2016	NA	169	NA	1	One
871	2017	NA	152	NA	1	One
872	2018	NA	153	NA	1	One
873	2019	NA	149	NA	1	One
874	2020	NA	164	NA	1	One
875	2021	NA	155	NA	1	One
876	2022	NA	151	NA	1	One
877	2023	NA	134	NA	1	One
878	2010	NA	71	NA	1	One
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880	2012	NA	59	NA	1	One
881	2013	NA	64	NA	1	One
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883	2015	NA	55	NA	1	One
884	2016	NA	55	NA	1	One
885	2017	NA	58	NA	1	One
886	2018	NA	53	NA	1	One
887	2019	NA	50	NA	1	One
888	2020	NA	47	NA	1	One
889	2021	NA	43	NA	1	One
890	2022	NA	45	NA	1	One
891	2023	NA	39	NA	1	One
892	2010	NA	160	NA	1	One
893	2011	NA	155	NA	1	One
894	2012	NA	157	NA	1	One
895	2013	NA	155	NA	1	One
896	2014	NA	161	NA	1	One
897	2015	NA	159	NA	1	One
898	2016	NA	166	NA	1	One
899	2017	NA	164	NA	1	One
900	2018	NA	166	NA	1	One
901	2019	NA	169	NA	1	One
902	2020	NA	235	NA	1	One

903	2021	NA	223	NA	1	One
904	2022	NA	175	NA	1	One
905	2010	NA	68	NA	1	One
906	2011	NA	65	NA	1	One
907	2012	NA	64	NA	1	One
908	2013	NA	60	NA	1	One
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911	2016	NA	56	NA	1	One
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914	2019	NA	50	NA	1	One
915	2020	NA	48	NA	1	One
916	2021	NA	49	NA	1	One
917	2022	NA	48	NA	1	One
918	2023	NA	49	NA	1	One
919	2010	NA	78	NA	1	One
920	2011	NA	77	NA	1	One
921	2012	NA	76	NA	1	One
922	2013	NA	71	NA	1	One
923	2014	NA	70	NA	1	One
924	2015	NA	68	NA	1	One
925	2016	NA	66	NA	1	One
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930	2012	NA	61	NA	1	One
931	2013	NA	56	NA	1	One
932	2014	NA	54	NA	1	One
933	2015	NA	52	NA	1	One
934	2016	NA	51	NA	1	One
935	2010	NA	126	NA	1	One
936	2011	NA	122	NA	1	One
937	2012	NA	120	NA	1	One
938	2013	NA	116	NA	1	One
939	2014	NA	107	NA	1	One
940	2015	NA	107	NA	1	One
941	2016	NA	106	NA	1	One
942	2017	NA	108	NA	1	One
943	2018	NA	106	NA	1	One
944	2019	NA	110	NA	1	One
945	2020	NA	119	NA	1	One
946	2021	NA	119	NA	1	One
947	2022	NA	114	NA	1	One
948	2010	NA	82	NA	1	One
949	2011	NA	79	NA	1	One
950	2012	NA	76	NA	1	One
951	2013	NA	74	NA	1	One
952	2014	NA	74	NA	1	One
953	2015	NA	72	NA	1	One

954	2016	NA	74	NA	1	One
955	2017	NA	71	NA	1	One
956	2018	NA	69	NA	1	One
957	2019	NA	65	NA	1	One
958	2022	NA	63	NA	1	One
959	2010	NA	167	NA	1	One
960	2012	NA	149	NA	1	One
961	2013	NA	150	NA	1	One
962	2014	NA	139	NA	1	One
963	2016	NA	137	NA	1	One
964	2017	NA	140	NA	1	One
965	2018	NA	134	NA	1	One
966	2019	NA	132	NA	1	One
967	2020	NA	136	NA	1	One
968	2021	NA	166	NA	1	One
969	2022	NA	142	NA	1	One
970	2023	NA	130	NA	1	One
971	2010	NA	81	NA	1	One
972	2011	NA	78	NA	1	One
973	2012	NA	75	NA	1	One
974	2013	NA	74	NA	1	One
975	2014	NA	69	NA	1	One
976	2015	NA	72	NA	1	One
977	2016	NA	65	NA	1	One
978	2017	NA	61	NA	1	One
979	2018	NA	64	NA	1	One
980	2019	NA	58	NA	1	One
981	2020	NA	57	NA	1	One
982	2021	NA	55	NA	1	One
983	2022	NA	53	NA	1	One
984	2023	NA	53	NA	1	One
985	2010	NA	64	NA	1	One
986	2011	NA	62	NA	1	One
987	2012	NA	60	NA	1	One
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991	2016	NA	56	NA	1	One
992	2017	NA	55	NA	1	One
993	2018	NA	54	NA	1	One
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997	2022	NA	51	NA	1	One
998	2023	NA	50	NA	1	One
999	2010	NA	64	NA	1	One
1000	2011	NA	63	NA	1	One
1001	2012	NA	62	NA	1	One
1002	2013	NA	61	NA	1	One
1003	2014	NA	57	NA	1	One
1004	2015	NA	59	NA	1	One

1005	2016	NA	55	NA	1	One
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1007	2018	NA	53	NA	1	One
1008	2019	NA	50	NA	1	One
1009	2020	NA	50	NA	1	One
1010	2021	NA	48	NA	1	One
1011	2022	NA	48	NA	1	One
1012	2023	NA	47	NA	1	One
1013	2010	NA	54	NA	1	One
1014	2011	NA	51	NA	1	One
1015	2012	NA	48	NA	1	One
1016	2013	NA	47	NA	1	One
1017	2014	NA	46	NA	1	One
1018	2015	NA	45	NA	1	One
1019	2016	NA	43	NA	1	One
1020	2017	NA	42	NA	1	One
1021	2018	NA	41	NA	1	One
1022	2019	NA	40	NA	1	One
1023	2020	NA	39	NA	1	One
1024	2021	NA	37	NA	1	One
1025	2022	NA	38	NA	1	One
1026	2023	NA	34	NA	1	One
1027	2010	NA	104	NA	1	One
1028	2011	NA	103	NA	1	One
1029	2012	NA	100	NA	1	One
1030	2013	NA	113	NA	1	One
1031	2014	NA	117	NA	1	One
1032	2015	NA	118	NA	1	One
1033	2016	NA	120	NA	1	One
1034	2017	NA	118	NA	1	One
1035	2018	NA	111	NA	1	One
1036	2019	NA	107	NA	1	One
1037	2020	NA	136	NA	1	One
1038	2021	NA	132	NA	1	One
1039	2022	NA	116	NA	1	One
1040	2023	NA	119	NA	1	One
1041	2010	NA	84	NA	1	One
1042	2011	NA	79	NA	1	One
1043	2012	NA	76	NA	1	One
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1046	2015	NA	74	NA	1	One
1047	2016	NA	74	NA	1	One
1048	2017	NA	73	NA	1	One
1049	2018	NA	75	NA	1	One
1050	2019	NA	71	NA	1	One
1051	2020	NA	71	NA	1	One
1052	2021	NA	71	NA	1	One
1053	2010	NA	96	NA	1	One
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1057	2014	NA	93	NA	1	One
1058	2015	NA	93	NA	1	One
1059	2016	NA	93	NA	1	One
1060	2017	NA	92	NA	1	One
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1070	2014	NA	136	NA	1	One
1071	2015	NA	141	NA	1	One
1072	2016	NA	149	NA	1	One
1073	2017	NA	141	NA	1	One
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1075	2019	NA	137	NA	1	One
1076	2020	NA	131	NA	1	One
1077	2021	NA	152	NA	1	One
1078	2022	NA	147	NA	1	One
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1085	2016	NA	148	NA	1	One
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1101	2020	NA	175	NA	1	One
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1125	2019	NA	96	NA	1	One
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```
health_df <- oecd_preventable

# Creating data frames from the ourdata package datasets
hdi_df <- hdi
imr_df <- imr
?imr

# Combining two data frames
?ourdata::combine
combined_df <- ourdata::combine(imr_df$name, hdi_df$country, imr_df$`deaths/1`, hdi_df$Human
                                col1 = "Country", col2 = "IMR", col3 = "HDI")
```

'data.frame': 181 obs. of 3 variables:

\$ Country: chr "Afghanistan" "Somalia" "Central African Republic" "Equatorial Guinea" ...

```
$ IMR      : num  101.3 83.6 80.5 77.4 71.2 ...
$ HDI      : num  0.496 0.404 0.414 0.674 0.467 0.419 0.416 0.388 0.493 0.419 ...
```

```
# or combining with dplyr
?inner_join
combined_df <- inner_join(imr_df, hdi_df, by = c("name" = "country")) %>%
  select(Country = name, IMR = `deaths/1`, HDI = HumanDevelopmentIndex_2024)

# Creating scatter plot
?plot
```

Help on topic 'plot' was found in the following packages:

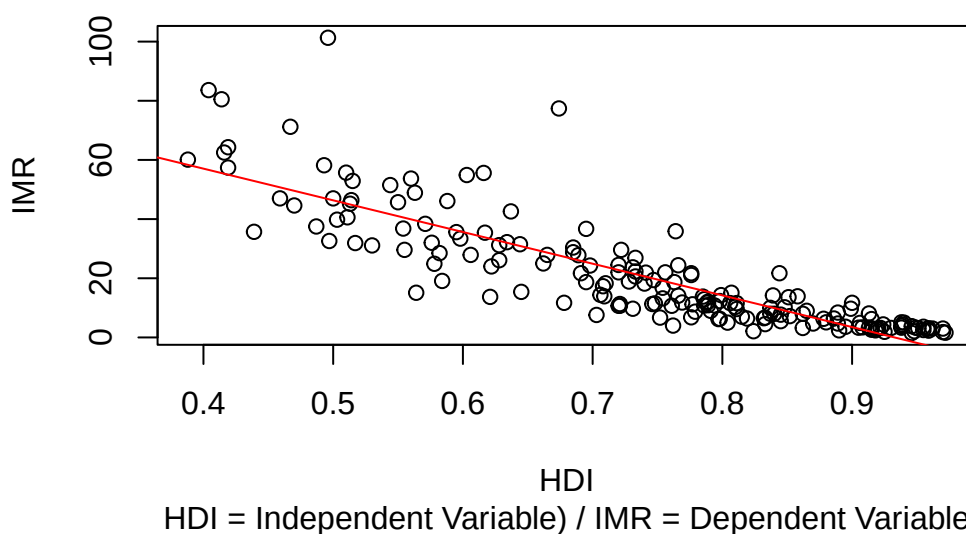
Package	Library
graphics	/opt/R/4.5.2/lib/R/library
base	/opt/R/4.5.2/lib/R/library

Using the first match ...

```
plot(combined_df$HDI, combined_df$IMR, main = "Influence of HDI (Human Development Index) on IMR (Infant Mortality Rate)",
      sub = "HDI = Independent Variable) / IMR = Dependent Variable", ylab = "IMR", xlab = "HDI")

# Adding regression line
model <- lm(combined_df$IMR ~ combined_df$HDI)
abline(model, col = "red")
```

Influence of HDI (Human Development Index) on IMR (Infant Mortality Rate)



```
# Showing summary of the model
summary(model)
```

```

Call:
lm(formula = combined_df$IMR ~ combined_df$HDI)

Residuals:
    Min       1Q   Median       3Q      Max
-24.387  -5.474  -0.017   4.470  54.530

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      99.893      3.623   27.57  <2e-16 ***
combined_df$HDI -107.103      4.775  -22.43  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.723 on 179 degrees of freedom
Multiple R-squared:  0.7376,    Adjusted R-squared:  0.7361
F-statistic: 503.2 on 1 and 179 DF,  p-value: < 2.2e-16

```

Conclusion and interpretation of the results:

The fitted linear model ($IMR \sim HDI$) shows a very strong, negative association: the HDI coefficient is -113.23 ($SE = 4.73$, $t = -23.92$, $p < 2e-16$).

This means that a one-unit increase in HDI (note: HDI ranges roughly 0–1) is associated with a predicted decrease in infant mortality of about 113 deaths per 1,000 live births.

Practical meaning Higher human development (better education, income and health components of HDI) is strongly associated with lower infant mortality.

Caveats and diagnostics to check Correlation causation: this is observational. Confounding (e.g., health spending, access to care, urbanization) could drive both HDI and IMR. Avoid causal claims without further analysis.

Summary HDI and IMR are strongly negatively associated: higher HDI is linked to substantially lower infant mortality (HDI explains $\sim 77\%$ of IMR variation here), but confirmatory causal analysis and diagnostic checks are needed before policy attribution.

5 Literature

All references for this course.

5.1 Essential Readings

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